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Individuals with anxiety and depression use atypical decision strategies in an uncertain world

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Abstract

Previous studies on reinforcement learning have identified three prominent phenomena: (1) individuals with anxiety or depression exhibit a reduced learning rate compared to healthy subjects; (2) learning rates may increase or decrease learning rate in environments with rapidly changing (i.e., volatile) or stable feedback conditions, a phenomenon termed *learning rate adaptation*; and (3) reduced learning rate adaptation is associated with several psychiatric disorders. In other words, multiple learning rate parameters are needed to account for behavioral differences across participant populations and volatility contexts in this flexible learning rate (FLR) model. Here, we propose an alternative explanation, suggesting that behavioral variation across participant populations and volatile contexts arises from the use of mixed decision strategies. To test this hypothesis, we constructed a mixture-of-strategies (MOS) model and used it to analyze the behaviors of 54 healthy controls and 32 patients with anxiety and depression in volatile reversal learning tasks. Compared to the FLR model, the MOS model can reproduce the three classic phenomena by using a single set of strategy preference parameters without introducing any learning rate differences. In addition, the MOS model can successfully account for several novel behavioral patterns that cannot be explained by the FLR model. Preferences towards different strategies also predict individual variations in symptom severity. These findings underscore the importance of considering mixed strategy use in human learning and decision making and suggest atypical strategy preference as a potential mechanism for learning deficits in psychiatric disorders.

eLife assessment

This study provides a novel and **valuable** alternative explanation for volatility-induced changes in choice behavior, commonly attributed to learning-rate adaptations. Through rigorous and comprehensive computational modeling of previously published data, the authors provide **convincing** support for the claim that apparent learning-rate adaptations may instead reflect a mixture of decision strategies. Furthermore, they demonstrate that differential weighting of the optimal decision strategy is predicted by psychopathology common to depression and anxiety. This work should be of interest to a wide range of scientists, including psychologists, neuroscientists, computer scientists, and clinicians.

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Introduction

Intelligent behavior requires the ability to adapt to an ever-changing environment. For example, foraging animals must be able to track the changing abundance or scarcity of food resources in different locations and at different timescales. Motor control demands the ability to control limbs that constantly vary in their dynamics (due to fatigue, injury, growth, etc.). Human competitors in games or sports of all varieties must be able to learn and adapt to the changing strategies of their opponents. To understand the mechanisms of these abilities, researchers have examined how (and how well) human agents can learn option values and track the dynamic changes in values in a *volatile reversal learning task* (Behrens et al., 2007). Unlike the traditional probabilistic reversal learning task where the reward probabilities of two options switch only once (Cools et al., 2002), this paradigm includes two volatility conditions (see **Fig. 1B**): the reward probabilities of the two options remain constant in one condition (i.e., the *stable* condition) and switch periodically in the other (i.e., the *volatile* condition).

Previous studies have often summarized human behaviors in this paradigm using the parameter of *learning rate*, which describes the efficiency with which current information is used to promote learning. These studies typically fit a specific learning rate to each context, resulting in different learning rate values for different contexts. Using this method, previous studies have reached two important conclusions. First, human participants are able to flexibly adapt to changes in environmental volatility, as evidenced by increasing and decreasing the learning rate in response to volatile and stable conditions. This observation is often referred to as the *learning rate adaptation* effect. Second, individuals with several psychiatric disorders, including anxiety and depression, have been found to have a reduced ability of learning rate adaptation in response to environmental volatility (Behrens et al., 2007; Browning et al., 2015; Gagne et al., 2020). This hallmark can also be indicative of atypical behaviors (Browning et al., 2015; Gagne et al., 2020), psychosis (Powers et al., 2017), and autism spectrum disorder (Lawson et al., 2017).

However, the current approach to understanding human behaviors using learning rates has two main limitations. First, the traditional approach increases the number of learning rates as the number of contexts increases, also increasing the risk of overfitting. Second, this approach implicitly assumes that learning rate differences can account for all behavioral differences between stable/volatile rewarding contexts and group differences between healthy controls and patients with psychiatric disorders. However, learning rate is not directly observable and is often estimated by model fitting, which limits its interpretability. The goal of this work is to offer an alternative explanation for human learning behaviors in volatile reversal learning tasks, moving

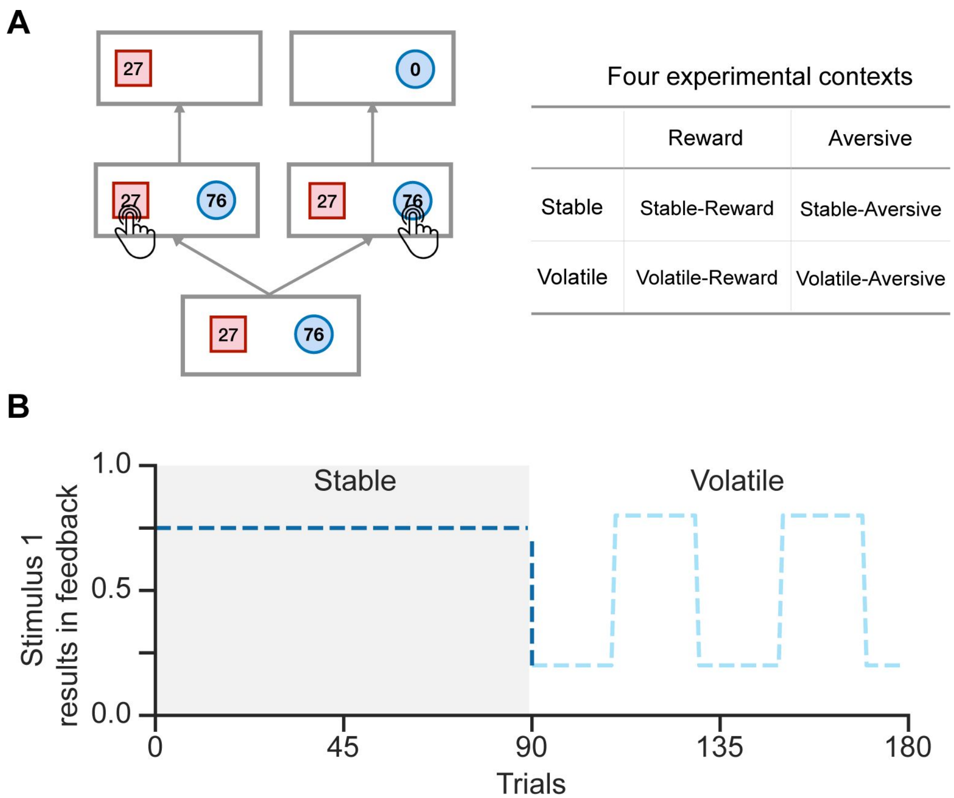

Figure 1.

Schematic diagram of the experimental task in Gagne et al. (2020).

A. In each trial, participants were presented with two stimuli associated with their potential feedback magnitude. They were instructed to choose one of the two stimuli to receive feedback, but only one stimulus would result in feedback. Participants were required to complete tasks across four experimental contexts.

B. Each run compromised 90 trials in the stable context and 90 trials in the volatile context. During the stable context, the true environmental probability remains unchanged, while in the volatile context, the probability flips every 20 trials.

beyond the traditional focus on learning rates. We hypothesize that the differences between stable/volatile contexts and between healthy/patient groups mainly arise from preferences for different decision strategies (Daw et al., 2011; Fan et al., 2023). We therefore built a novel mixture-of-strategies (MOS) model which postulates that an observer makes decisions by combining three strategies that balance reward and cognitive resources (Gershman et al., 2015; Griffiths et al., 2015). First, we consider the most rewarding strategy, *Expected Utility* (EU), which guides decision-making based on the expected utility of each option (calculated as probability multiplied by reward magnitude) (Von Neumann & Morgenstern, 1947). This EU strategy yields the maximum amount of reward, but the utility calculation itself consumes considerable cognitive resources. Alternatively, humans may choose simpler strategies, e.g., the *magnitude-oriented* (MO) strategy, in which only the reward magnitude was considered during the decision process, and the *habitual* (HA) strategy, in which people simply repeat decisions frequently made in the past regardless of reward magnitude (Wood & Runger, 2016). We use the preference for these decision strategies to roughly estimate participants' decision style in the volatile reversal task.

In this study, we apply and examine the MOS model on a dataset previously reported by Gagne et al. (2020) and demonstrate its ability to explain the impaired learning behaviors of individuals diagnosed with depression and anxiety. First, we show that depression and anxiety patients exhibit three signature behavioral patterns indicative of inferior task performance. The MOS model not only qualitatively captures all three behavioral patterns but also quantitatively provides a better fit to the behavioral data than previous models. We then revisit the classical learning rate adaptation theory and show that strategy preference readily accounts for two key learning rate adaptation effects observed in prior research. Our work presents an alternative explanation for the effects of environmental volatility on human learning and highlights the importance of understanding atypical patient behaviors through the lens of decision-making strategies rather than solely focusing on learning rate.

Methods and Materials

Datasets

We focused on the data from Experiment 1 reported in Gagne et al. (2020). The data is publicly available via (*https://osf.io/8mzuj/*). The original study included data from two experiments. The data from Experiment 2 was not used here because it was implemented on Amazon's Mechanical Turk with no information about the participants' clinical diagnoses. Here, we provide critical information about Experiment 1 (also see Gagne, et al. (2020) for more technical details).

Participants

Eighty-six participants took part in this experiment. The pool includes 20 patients with a major depressive disorder (MDD), 12 patients with a generalized anxiety disorder (GAD), and 54 healthy control participants. The diagnosis was made through a phone screen, an in-person screening session, and the structured clinical interview following DSM-IV-TR (SCID) in 20 MDD patients, 12 GAD patients, and 20 healthy control participants. The remaining 30 healthy control participants were recruited without SCID. In this article, we grouped the MDD and GAD individuals into a patient (PAT) group and the remaining 54 participants into a healthy control (HC) group. The detailed difference between MDD and GAD is not the focus of this paper. We will show later that the general factor behind MDD and GAD is the only factor that predicts learning behavior (see next section for details), a similar result reported in the original study (Gagne et al., 2020).

Clinical measures

The severity of anxiety and depression in all participants was measured by several standard clinical questionnaires, including the Spielberger State-Trait Anxiety Inventory (STAI form Y; Spielberger CD, 1983), the Beck Depression Inventory (BDI; Beck et al., 1961), the Mood and Anxiety Symptoms Questionnaire (MASQ; Clark & Watson, 1991; Watson & Clark, 1991), the Penn State Worry Questionnaire (Meyer et al., 1990), the Center for Epidemiologic Studies Depression Scale (CESD; Radloff, 2016), and the Eysenck Personality Questionnaire (EPQ; Eysenck & Eysenck, 1975). An exploratory bifactor analysis was then applied to item-level responses in all questionnaires to disentangle the variance that is common to GAD and MDD or unique to each. The results of this analysis summarized participants' symptoms into three orthogonal factors: a general factor (g) explaining the common symptoms, a depression-specific factor (f1), and an anxiety-specific factor (f2), which are all included in the public dataset. Similar to the original study, here we focused on the general factor (*g* score) to indicate the participants' severity of their psychiatric symptoms.

Stimuli and behavioral task

In a volatile reversal learning task (**Fig. 1A**), participants on each trial were instructed to choose between two stimuli, represented by different shapes, in order to receive feedback. The locations of the two shapes were counterbalanced across trials. The potential amount of feedback (referred to as feedback magnitude) was presented together with the stimuli. Only one of the two stimuli was associated with actual feedback (0 for the other one). The feedback magnitude, ranged between 1-99, is sampled uniformly and independently for each shape from trial to trial. Actual feedback was delivered only if the stimulus associated with feedback was chosen; otherwise, a number "0" was displayed on the screen, signifying that the chosen stimulus returns nothing.

Participants was supposed to complete this learning and decision-making task in four experimental contexts (**Fig. 1A**), two feedback contexts (reward or aversive) × two volatility contexts (stable or volatile). Participants received points in the reward context and an electric shock in the aversive context. The reward points in the reward context were converted into a monetary bonus by the end of the task, ranging from £0 to £10. In the stable context, the dominant stimulus (i.e., a certain stimulus induces the feedback with a higher probability) provided a feedback with a fixed probability of 0.75, while the other one yielded a feedback with a probability of 0.25. In the volatile context, the dominant stimulus's feedback probability was 0.8, but the dominant stimulus switched between the two every 20 trials. Hence, this design required participants to actively learn and infer the changing stimulus-feedback contingency in the volatile context.

Each participant was instructed to complete two runs of the volatile reversal learning task, one in the reward context and the other in the aversive context. Each run consisted of 180 trials, with 90 trials in the stable context and 90 in the volatile context (**Fig. 1B**). No additional hints were provided about the transition from one context to another, thereby participant needs to infer the current context on their own. A total of 79 participants completed tasks in both feedback contexts. 4 participants only completed the task in the reward context, while 3 participants only completed the aversive task.

Computational Modeling

We first introduce our notation system. We denote each stimulus s as one of two possible states $s \in \{s_1, s_2\}$. The labeled feedback magnitude (i.e., reward points or shock intensity) of the stimulus is $m(s)$, and the feedback probability is $\psi(s)$. Following the convention in reinforcement learning

(Sutton & Barto, 2018), we presume that the decision is made from a policy π that maps the observed magnitudes m and currently maintained feedback probabilities ψ to a distribution over stimuli, $\pi(s|m, \psi)$.

In a volatile reversal learning task, each participant in the experiment must resolve two fundamental challenges: 1) decision-making, determining an action to maximize benefit; and 2) learning, figuring out the untold feedback probability via their interaction with the environment. Here, we introduce four families of models that all utilize the same reinforcement learning method in learning feedback probability but differ in how they construct their policies for decision-making. First, the mixture-of-strategies (MOS) model, the target model proposed in this paper, utilizes a decision-making policy consisting of a mixture of three strategies: expected utility (EU), magnitude-oriented (MO), and habitual (HA). Second, the flexible-learning-rate (FLR) model, reported as the best model in Gagne et al., (2020), selects stimuli with higher values. Stimulus value was estimated by a linear combination of differences in feedback probability, (non-linear) feedback magnitude, and the stimuli's consistency with habitual behaviors. Third, the risksensitive (RS) model, adopted from Behrens et al., (2007) and Browning et al., (2015), utilizes the EU strategy in decision-making and considers a subjective distortion of the learned feedback probability when calculating expected value. Finally, the Pearce-Hall (PH) model, equipped with a built-in learning rate adaptation mechanism, utilizes the EU strategy for decision-making.

Notably, the MOS model, as the core contribution of this study, posits that behavioral differences across the two participant groups and stable/volatile contexts is due to different weightings of multiple decision strategies. In contrast, the other three models posit that the behavioral differences mainly arise via different learning rates between groups and contexts.

The Mixture-of-Strategies (MOS) model

The key signature of the MOS model is that its policy consists of a mixture of three strategies: EU, MO, and HA. Among many possible variants of the MOS models, this particular three-strategy configuration was chosen as the representative model because it best accounts for human behavioral data (***Supplemental Note 1***).

The EU strategy postulates that human agents rationally calculate the value of each stimulus and use the softmax rule to select an action. In this case, the value of a stimulus should be its expected utility: $m(s)\psi(s)$. The probability of choosing a stimulus s thus follows a softmax function.

$$\pi_{\mathrm{EU}}(s|\psi, m) = \frac{\exp\big(\beta\psi(s)m(s)\big)}{\sum_{s'}\exp\big(\beta\psi(s')m(s')\big)} \tag{1}$$

where β is the inverse temperature. For simplicity, we rewrite Eq. 1 in the following form:

$$\pi_{\mathrm{EU}}(s|\psi, m) = \mathrm{softmax}\big(\beta\psi(s)m(s)\big) \tag{2}$$

Different from the EU strategy, the MO strategy postulates that observers only focus on feedback magnitude $m(s)$, disregarding feedback probability $\psi(s)$. This is certainly an irrational strategy but more economical in terms of cognitive efforts. Feedback magnitudes are explicitly shown with the stimuli in each trial and readily available for related cognitive computation. But feedback probability, as a latent variable, requires trial-by-trial learning and inference, which is more cognitively demanding. The MO strategy is defined as,

$$\pi_{\mathrm{MO}}(s|m) = \mathrm{softmax}\big(\beta m(s)\big) \tag{3}$$

Unlike EU and MO, the HA strategy depends on neither feedback magnitude $\psi(s)$ nor feedback probability -0(s). The HA strategy reflects the tendency to repeat previous frequent choices. This tendency reflects the habit of choosing a stimulus, a phenomenon called perseveration in literature (Gershman, 2020; Wood & Runger, 2016). For example, if an agent chooses stimulus 1 more often in past trials, she will form a preference for stimulus 1 in future trials. We constructed it as a Bernoulli distribution over the two stimuli $\pi_{\mathrm{HA}}(s)$. The trial-by-trial update rule of $\pi_{\mathrm{HA}}(s)$ will be detailed in Eqs. 5–6 below.

We implemented the hybrid policy of a linear mixture of the three strategies following the methods used in Daw et al. (2011),

$$\pi(s|\psi, m, \pi_{\mathrm{HA}}) = w_{EU}\pi_{\mathrm{EU}}(s|\psi, m) + w_{\mathrm{MO}}\pi_{\mathrm{MO}}(s|m) + w_{\mathrm{HA}}\pi_{\mathrm{HA}}(s) \tag{4}$$

where w_{EU}, w_{MO}, and w_{HA} are the weighting parameters of each strategy. The three weighting parameters should be summed to 1, i.e., $w_{\mathrm{EU}} + w_{\mathrm{MO}} + w_{\mathrm{HA}} = 1$. We can thus describe the policy an observer adopted just by examining the weighting parameters. Formulating the hybrid model in this way improves the interpretability of the weighting parameters because all three decision strategies are constructed in a Bernoulli format.

Next, we solve the challenge of probabilistic learning. Two distributions — the feedback probability and the habit— are learned and updated in a trial-by-trial fashion. We updated the feedback probability in a Rescorla-Wagner format (Rescorla, 1972):

$$\begin{aligned} \psi(s_1) &= \psi(s_1) + \alpha_\psi\big(O(s_1) - \psi(s_1)\big) \\ \psi(s_2) &= 1 - \psi(s_1) \end{aligned} \tag{5}$$

where α_ψ is the learning rate for feedback probability. $O(\cdot)$ is an indicator function that returns 1 at the true feedback stimulus and 0 otherwise. To keep consistent with Gagne et al., (2020), we also explored the valence-specific learning rate,

$$\alpha_\psi = \begin{cases} \alpha_{\psi+}, & \text{for } (O(s_1) - \psi(s_1)) > 0 \\ \alpha_{\psi-}, & \text{for } (O(s_1) - \psi(s_1)) < 0 \end{cases} \tag{6}$$

$\alpha_{\psi+}$ is the learning rate for better-than-expected outcome, and $\alpha_{\psi-}$ for worse-than-expected outcome. It is important to note that Eq. 6 was only applied to the reward context, and the definitions of "better-than-expected" and "worse-than-expected" should change accordingly in the aversive context, where we defined $\alpha_{\psi+}$ for $(O(s_1) - \psi(s_1)) < 0$ and $\alpha_{\psi-}$ for $(O(s_1) - \alpha(s_1)) > 0$.

The habit component is updated in a similar manner.

$$\begin{aligned} \pi_{\mathrm{HA}}(s_1) &= \pi_{\mathrm{HA}}(s_1) + \alpha_{\mathrm{HA}}\big(A(s_1) - \pi_{\mathrm{HA}}(s_1)\big) \\ \pi_{\mathrm{HA}}(s_2) &= 1 - \pi_{\mathrm{HA}}(s_1) \end{aligned} \tag{7}$$

where α_{HA} is the learning rate for the habitual strategy. $A(\cdot)$ is also an indicator function that returns 1 for the stimulus chosen at the current trial. Intuitively, the stimulus chosen more often will results in a higher π_{HA} for subsequent trials.

We developed two variants of the MOS model: a context-free and a context-dependent variant. The context-free MOS6 has a total of 6 free parameters $\xi = \{\beta, \alpha_{\mathrm{HA}}, \alpha_\psi, w_{\mathrm{EU}}, w_{\mathrm{MO}}, w_{\mathrm{HA}}\}$. This variant does not include the design of value-specific learning rate. The context dependent variant MOS22 has a total of 22 free parameters. Among them β and α_{HA} are context-free parameters that were held the same across all contexts. Parameters $\{\alpha_{\psi+}, \alpha_{\psi-}, w_{\mathrm{EU}}, w_{\mathrm{MO}}, w_{\mathrm{HA}}\}$ are context-dependent parameters that should be fitted independently to each context.

We fit the context-dependent parameters to each context following a 2 (reward/aversive) × 2(stable/volatile) factorial structure (**Fig. 1A**). Specifically, the five context-dependent parameters, positive learning rate parameter $\alpha_{\psi+}$, negative learning rate parameter $\alpha_{\psi-}$, and three strategies weights w_{EU}, w_{MO}, w_{HA} were fit separately to each context. The remaining two parameters $\{\beta, \alpha_{\mathrm{HA}}\}$ were held constant across all four experimental contexts for each participant. Thus, there were 22 free parameters (5 context-dependent parameters × 4 conditions + 2 contextfree parameters) of the MOS model in each participant.

The Flexible Learning Rate (FLR) model

The FLR model refers to Model 11 (i.e., the best-fitting model) in Gagne et al. (2020). Here, we describe the FLR model using the same notation system with the published paper, slightly different from the notations in the MOS model. The FLR model models the probability of selecting the stimulus 1 as,

$$\pi(s_1|v, \pi_{\mathrm{HA}}) = \frac{1}{1+\exp(-\beta v - \beta_{\mathrm{HA}}[\pi_{\mathrm{HA}}(s_1) - \pi_{\mathrm{HA}}(s_2)])} \tag{8}$$

where β and β_{HA} are the inverse temperature parameters of the value of the stimulus 1 and the HA strategy, respectively. The value of stimulus 1 represents the advantage of s_1 over s_2,

$$v = \lambda[\psi(s_1) - \psi(s_2)] + (1-\lambda)\mathrm{sign}\big(m(s_1) - m(s_2)\big)|m(s_1) - m(s_2)|^r \tag{9}$$

where λ is the weighting parameter balancing the two terms. The first term $\psi(s_1) - \psi(s_2)$ indicates the feedback probability difference between the two options. The second term, $\mathrm{sign}(m(s_1) - m(s_2))|m(s_1) - m(s_2)|^r$, indicates the feedback magnitude differences scaled by a non-linear factor r. Intuitively, the value v of s_1 can be understood as the weighted sum of the feedback probability differences and the feedback magnitude difference.

During the learning stage, the FLR model learns the feedback probability using the same equations in the MOS model (Eqs. 5–6). The context-free variant FLR6 has 6 free parameters $\psi = \{\alpha_{\mathrm{HA}}, r, \beta_{\mathrm{HA}}, \beta_{\psi}, \beta, \lambda\}$. The context-dependent variant FLR22 considers $\{\alpha_{\mathrm{HA}}, r\}$as contextfree parameters and $\{\beta_{\mathrm{HA}}, \alpha^{\psi+}, \alpha_{\psi-}, \beta, \lambda\}$ as context-dependent parameters, resulting in a total of 22 free parameters.

The Risk-Sensitive (RS) model

We adopted the RS model from Behrens, et al. (2007). The RS model assumes that participants apply the EU strategy but with a subjectively distorted feedback probability $\tilde{\psi}(s_1)$,

$$\pi\big(s_1|\tilde{\psi}, m\big) = \frac{1}{1+\exp(-\beta[\tilde{\psi}(s_1)m(s_1) - \tilde{\psi}(s_2)m(s_2)])} \tag{10}$$

where β is the inverse temperature. The distorted probability is calculated by,

$$\tilde{\psi}(s_1) = \max[\min[(\gamma(\psi(s_1) - 0.5) + 0.5, 1], 0]$$
$$\tilde{\psi}(s_2) = 1 - \tilde{\psi}(s_1) \tag{11}$$

where the γ indicates participants' risk sensitivity. When $\gamma = 1$, a participant has an unbiased risk balance. $\gamma < 1$ and $\gamma > 1$ indicate risk-seeking and risk-aversive tendencies, respectively.

The RS model learns the feedback probability in the same way as the MOS and FLR models (i.e., Eq. 5). The model did not include the HA strategy. The context-free variant RS3 has a total of 3 free parameters $\xi = \{\beta, \alpha_\psi, \gamma\}$. The context-dependent variant RS13 considers $\{\beta\}$ as a context-free parameter and $\{\alpha_{\psi+}, \alpha_{\psi-}, \gamma\}$ as context-dependent parameters, resulting in a total of 13 free parameters.

The Pearce-Hall (PH) model

To explicitly incorporate a learning rate adaptation mechanism, we adopt the PH model from Pearce and Hall (1980). This model proposes an adaptive learning rate, as outlined in Eq. 5,

$$\psi(s_1) = \psi(s_1) + k\alpha_\psi\big(O(s_1) - \psi(s_1)\big)$$
$$\psi(s_2) = 1 - \psi(s_1) \tag{12}$$

where k is a scale factor of the learning rate. Each trial the learning rate is updated in accordance with the absolute prediction error,

$$\alpha_\psi = \alpha_\psi + \eta\big(|O(s_1) - \psi(s_1)| - \alpha_\psi\big) \tag{13}$$

where η is the step size for the learning rate. We have no knowledge of participants' learning rate values before the experiment, so we need to also fit the initial learning rate value, α_ψ^0. The PH model generate a choice through the EU strategy:

$$\pi(s_1|\psi, m) = \frac{1}{1 + \exp(-\beta[\psi(s_1)m(s_1) - \psi(s_2)m(s_2)])} \tag{14}$$

The context-free variant PH4 has a total of 4 free parameters $\xi = \{\alpha_\psi^0, k, \eta, \beta\}$. The contextdependent variant PH17 considers $\{\alpha_\psi^0\}$ as a context-free parameter and $\{k_+, k_-, \eta, \gamma\}$ as contextdependent parameters, resulting in a total of 17 free parameters.

Model fitting

Parameters were estimated for each participant via the maximum a posteriori (MAP) method. The objective function to maximize is:

$$\max_\xi \sum_{i=1}^{N} \log L(s_i|m_i, O_i, \mathrm{M}, \xi) + \log p(\xi) \tag{15}$$

where ξ means the model free parameters. M is the model and N refers to the number of trials of the participant's behavioral data. m_i, O_i, and s_i are the presented magnitude, true feedback probability, and participants' responses recorded in each trial.

Parameter estimation was performed using the *Broyden-Fletcher-Goldfarb-Shanno (BFGS)* algorithm in the *scipy.optimize* module in Python. This algorithm provides an approximation of the inverse Hessian matrix for the parameter, a critical component that can be employed in Bayesian model selection (Rigoux et al., 2014). In order to use the BFGS algorithm, we reparametrized the model, thereby transforming the original fitting problem into an unconstrained optimization problem. We carefully tuned the parameter priors to ensure that they had little impact on the fitting results (***Supplemental Note 2***). For each participant, we ran the optimization with 40 randomly chosen initial parameters to avoid local minima.

Importantly, to fit the weighting parameters (w_{EU}, w_{MO}, w_{HA}) and ensure them summed to 1, we parameterized the weighting parameters as outputs of a softmax function,

$$w_i = \mathrm{softmax}(\lambda_i) \qquad \forall i \in \{\mathrm{EU}, \mathrm{MO}, \mathrm{HA}\} \tag{16}$$

and fit the logits λ_i of the weights. All logits were assumed to be normally distributed with a prior N(0,10). In the result section, we used both (w_{EU}, w_{MO}, w_{HA}) and (λ_{EU}, λ_{MO}, λ_{HA}) to represent participants' strategy preferences. Some of the statistical analyses were performed only on (λ_{EU}, λ_{MO}, λ_{HA}) because they are normally distributed.

Simulation details

Simulate to understand the three strategies

We run simulations to understand the effects of the three strategies on hit rate, hit rate difference, and learning curve. We first used the MOS6 to simulate the learning behaviors of the healthy control group in 100 independent experiments. The parameters were set as β = 10.803, α_{HA} = 0.423, α_{ψ} = 0.473, α_{EU} = 1.138, α_{MO} = -1.547, α_{HA} = 0.686, where the first three parameters represent the median across both groups and the latter three weighting parameters are the median across healthy controls. Each simulated experiment consists of 2 runs, one showing a stable context first and then a volatile context, and vice versa in the other run. This approach results in a total of 200 runs for the healthy control group. The task sequences were randomly generated using the same design Gagne et al. (2020) used for data collection. Similarly, we repeated all the simulation procedures for the patient group, except that the parameters were set to β = 10.803, β_{HA} = 0.423, α_{ψ} = 0.473, λ_{EU} = 0.515, λ_{MO} = -0.220, λ_{HA} = 0.094. Note that we used identical {β, α_{HA}, α_{ψ}} in both groups and only varied {λ_{EU}, λ_{MO}, λ_{HA}} as the median across the patient participants. We used π_{EU} $(s|\psi, m)$, π_{MO} $(s|m)$, and π_{HA} (s) to evaluate the task performance associated with each strategy (e.g., **Fig. 4B-D**). We did not run each strategy completely independently because the HA strategy alone cannot complete the task without learning from decisions previously made by the EU strategy.

Simulate to explain learning rate adaptation using MOS6

In one simulated experiment, we sampled the four task sequences from the real data. We simulated 20 experiments with the parameters of β = 10.803, β_{HA} = 0.423, β_{ψ} = 0.473, w_{EU} = 0.60, w_{MO} = 0.15, w_{HA} = 0.25 to mimic the behavior of the healthy control participants. The first three are the median of the fitted parameters across all participants; the latter three were chosen to approximate the strategy preferences of real healthy control participants (**Figure 4A**). Similarly, we also simulated 20 experiments for the patient group with the identical values of β, α_{HA}, and α_{ψ}, but different strategy preferences w_{EU} = 0.15, w_{MO}= 0.60, w_{HA} = 0.25. In other words, the only difference in the parameters of the two groups is the switched w_{EU} and w_{MO}. We then fitted the FLR22 to the behavioral data generated by the MOS6 and examined the learning rate differences across groups and volatile contexts (**Fig. 6**).

Results

Atypical behavioral patterns in the MDD and GAD patients

By analyzing the behavioral data, we demonstrate that patients with MDD and GAD exhibit three key behavioral patterns as compared to the healthy controls.

First, patients achieved a significantly lower hit rate (averaged across stable and volatile contexts) as compared to the healthy controls (**Fig. 2A**; $t(70.541) = 3.326$, $p = 0.001$, Cohen's d = 0.723). The hit rate refers to the accuracy of a participant in choosing the correct stimulus throughout the task. Specifically, the correct stimulus is the one that yields reward points in the reward or avoids electric shocks the aversive context.

Second, we observed two atypical features of learning curves in the patient group (**Fig. 2B**). The patients' learning curves took more trials to converge to an asymptote (i.e., seemingly slower learning). Additionally, there was a larger apparent deviation (**Fig. 2B**, blue arrows) from the true feedback probability. The apparent deviation indicates that the learning curve of the patient group could never converge to the true feedback probability even given a sufficient number of trials in the stable context.

Third, aside from the lower learning rate and atypical learning curves that indicating inferior performance of the patient group, we further discovered a reduced hit rate difference within the patient group (**Fig. 2C**, $t(55.648) = 2.038$, $p = 0.046$, Cohen's d = 0.478). Interestingly, this hit rate difference is marginally associated with the severity of participants' symptom ($r(86) = -0.194$,p = 0.074), as measured using the bifactor analysis reported in Gagne et al. (2020). This analysis decomposes symptoms into specific factors for anxiety and depression, with the *g-score* representing the common symptoms between them. The hit rate difference across volatile/stable contexts may be due to the setting of true probability (0.8 in the volatile context and 0.75 in the stable context).

The mixture-of-strategies model captures group differences in learning behaviors

To understand patients' behavioral patterns, we fitted and compared a total of eight computational models. The two Mixture-of-Strategies (MOS) models, MOS6 and MOS22, are proposed in our study. These models posit that behavioral differences across the two participant groups and between stable/volatile contexts are due to different weightings of multiple decision strategies: expected utility (EU), magnitude-oriented (MO), and habitual (HA). Both models have identical update rules; however, MOS22, the context-dependent variant, fits a separate set of parameters to each experimental context, whereas MOS6, the context-free variant, uses one set of parameters for all contexts (**Table 1**). In contrast, the Flexible Learning Rate (FLR) models—the context-free FLR6 and the context-dependent FLR22—hypothesize that behavioral differences between groups and contexts primarily arise from different learning rates, known as learning rate adaptation. These models, reported as the best models in Gagne et al., (2020), select stimuli with higher values, estimated by a linear combination of differences in feedback probability, (non-linear) feedback magnitude, and the stimuli's consistency with habitual behaviors. The Risk-Sensitive (RS) models (RS3 and RS13), adopted from Behrens et al. (2007) and Browning et al. (2015), share the same hypothesis about human behavioral differences. These models utilize the EU strategy in decision-making and consider a subjective distortion of the learned feedback

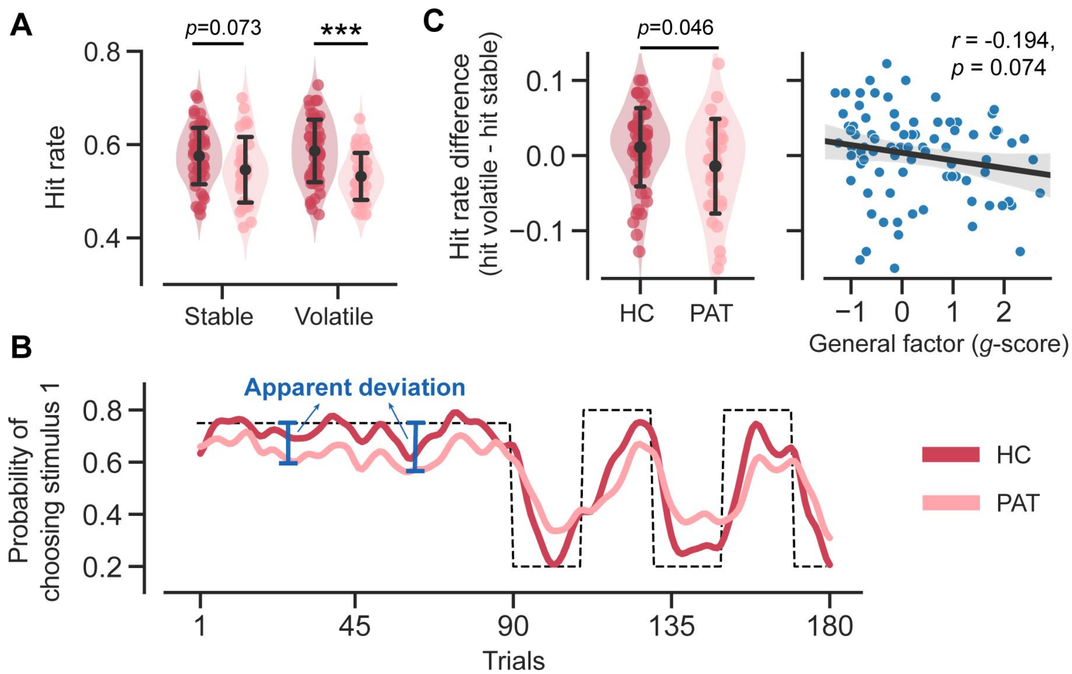

Figure 2.

Task performance comparison between healthy control participants and patients diagnosed with MDD and GAD. Significance symbols: *: *p* < 0.05; **: *p* < 0.01; *: *p* < 0.001; *n.s.*: non-significant. Abbreviations: HC-healthy controls, PAT-patients.**

A. Comparison of hit rates for healthy controls and patients in stable and volatile contexts. Error bars represent the standard deviation of the estimated mean across 86 participants.

B. Learning curves for healthy controls and patients throughout the learning process. The dash line represents the exemplar feedback probability sequence. For runs that do not follow this exemplar sequence (e.g., starting with volatile then moving to stable conditions), responses were converted to match the exemplar sequence. The learning curves for both groups were then generated by averaging these converted responses across participants within each group. For better visualization, these curves were then smoothed using a Gaussian kernel with a standard deviation of 2 trials. The blue arrows indicate the apparent deviation between the true feedback probability and the patients' asymptotic performance.

C. Hit rate differences for healthy controls and patients, and its relationship with participants' symptom severities. Error bars respectively represent the standard deviation of the estimated mean across 54 healthy controls and 32 patients.

probability when calculating expected value. To further investigate the hypothesis regarding differences in learning rates, we tested a family of models with a built-in adaptive learning rate, known as the Pearce-Hall models (PH4, PH17).

The model fitting reveals that the MOS models accurately account for human behaviors. MOS6 and MOS22 were the best-fitting models in terms of Bayesian Information Criterion (BIC; Schwarz, 1978) and Akaike Information Criterion (AIC; Akaike, 1974), respectively (**Fig. 3A**). The group-level Bayesian model comparison (Rigoux et al., 2014) further supports MOS6 as the best-fitting model (**Fig. 3B**). These model comparisons highlight that the MOS models outperform the other three families of models supporting for the learning adaptation account, suggesting that behavioral variations might not be fully captured by learning rate adaptations alone.

The MOS models can not only quantitatively better capture the data but also reproduce the three key behavioral differences between the groups effectively. The MOS models reproduce the lower hit rate (**Fig. 3C**), reduced hit rate difference (**Fig. 3D**), and slower learning curves with apparent deviations (**Fig. 3E**) observed in the patient group, whereas the FLR models struggle to produce all these effects. See ***Supplemental Note 3*** for the behavioral patterns for all models.

In summary, we conclude that the MOS models best account for human behavioral data both qualitatively and quantitatively. In the following sections, we will analyze the fitted parameters of the MOS models to interpret the atypical behavioral patterns of the patient group.

MDD and GAD patients favor simpler decision strategies

We first focused on the fitted parameters of the MOS6 model. We compared the weight parameters (w_{EU}, w_{MO}, w_{HA}) across groups and conducted statistical tests on their logits (λ_{EU}, λ_{MO}, λ_{HA}). The patient group showed a ~37% preference towards the EU strategy, which is significantly weaker than the ~50% preference in healthy controls (healthy controls' λ: M = 0.991, SD = 1.416; patients' λ: M = 0.196, SD = 1.736; $t(54.948) = 2.162$, $p = 0.035$, Cohen's d = 0.509; **Fig. 4A**). Meanwhile, the patients exhibited a weaker preference (~27%) for the HA strategy compared to healthy controls (~36%) (healthy controls' λ: M = 0.657, SD = 1.313; patients' λ: M = -0.162, SD = 1.561; $t(56.311) = 2.455$, $p = 0.017$, Cohen's d = 0.574), but a stronger preference for the MO strategy (14% vs. 36%; healthy controls' λ: M = -1.647, SD = 1.930; patients' λ: M = -0.034, SD = 2.091; $t(63.746) = -3.510$, $p = 0.001$, Cohen's d = 0.801). Most importantly, we also examined the learning rate parameter in the MOS6 but found no group differences ($t(68.692) = 0.690$, $p = 0.493$, Cohen's d = 0.151). These results strongly suggest that the differences in decision strategy preferences can account for the learning behaviors in the two groups without necessitating any differences in learning rate *per se*.

The MOS6 assumes no parameter differences across the four contexts, which may dilute the group differences in learning rate. We further analyzed the MOS22, which explicitly estimates different set of weighting and learning rate parameters in different contexts (i.e., contextdependent), and found a consistent conclusion about participants' strategy preferences. We first conducted three separate 2 × 2 × 2 ANOVAs, each setting the logit of a weighting parameter (λ_{EU}, λ_{MO}, λ_{HA}) as the dependent variable, and participant groups (healthy control/patient) as the between-subject variable, and volatile contexts (stable/volatile), and feedback contexts (reward/aversive) as within-subject variables. We again found a weaker preference for EU ($F(1, 80) = 13.537$, $p < 0.001$, $\eta^2 = 0.084$) and a stronger preference for MO ($F(1, 80) = 7.791$,$p = 0.009$, $\eta^2 = 0.046$) in the patient group. However, unlike the MOS6, the MOS22 revealed no significant group difference was observed in the HA strategy ($F(1, 80) = 0.020$, $p = 0.887$, $\eta^2 < 0.001$), and a significant a stronger preference for EU under the reward context (Bonferroni-$t = 2.243$, $p = 0.028$, Cohen's $d = 0.209$. This suggests a possible confounding between the EU and HA strategies. Next, we examined the learning rates of the MOS22. A 2 × 2 × 2 × 2 ANOVA was performed with the (log) learning rate parameter as the dependent variable, outcome valence (better/worse than expectation) as a within-subject variable in addition to the three independent variables (group/volatile context/feedback context) as introduced above. We again found no significant difference between patients and healthy controls

Model	Context-free parameters	Context-dependent parameters
MOS6	$\beta, \alpha_{\mathrm{HA}}, \alpha_{\psi}, w_{\mathrm{EU}}, w_{\mathrm{MO}}, w_{\mathrm{HA}}$	
MOS22	$\beta, \alpha_{\mathrm{HA}}$	$\alpha_{\psi+}, \alpha_{\psi-}, w_{\mathrm{EU}}, w_{\mathrm{MO}}, w_{\mathrm{HA}}$
FLR6	$\alpha_{\mathrm{HA}}, r, \beta_{\mathrm{HA}}, \alpha_{\psi}, \beta, \lambda$	
FLR22	α_{HA}, r	$\beta_{\mathrm{HA}}, \alpha_{\psi+}, \alpha_{\psi-}, \beta, \lambda$
RS3	$\beta, \alpha_{\psi}, \gamma$	
RS13	β	$\alpha_{\psi+}, \alpha_{\psi-}, \gamma$
PH4	$\alpha_{\psi}^{0}, k, \eta, \beta$	
PH17	α_{ψ}^{0}	$k_{+}, k_{-}, \eta, \gamma$

Table 1

Model’s parameters

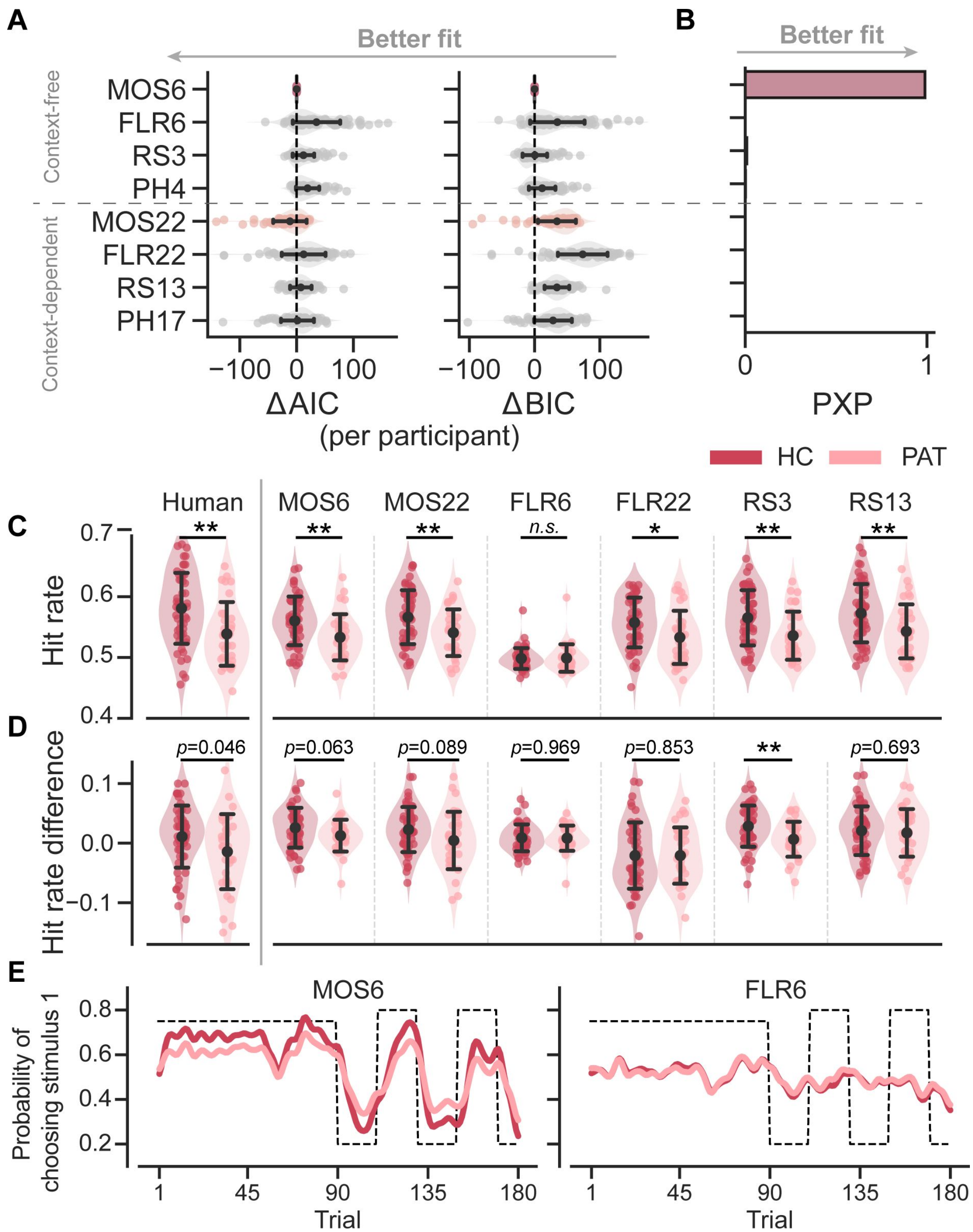

Figure 3.

Models' quantitative and qualitative fit to human behavioral data. Significance symbols: *: p < 0.05; **: p < 0.01; ***: p < 0.001; *n.s.*: non-significant. Abbreviations: HC - healthy controls, PAT - patients.

A. Relative performance of models compared to the MOS6 model, as measured by Akaike Information Criterion (AIC), and Bayesian Information Criterion (BIC). Each dot represents a model's fit for an individual participant, with error bars showing the standard deviation of the estimated mean across 86 participants.

B. Group-level Bayesian model selection as indicated by Protected Exceedance Probability (PXP).

C-E. Models' predicted hit rate (C), hit rate differences (D), and learning curves (E) for healthy controls and patients, respectively. Error bars denote the standard deviation of the estimated mean across 54 healthy controls and 32 patients, respectively.

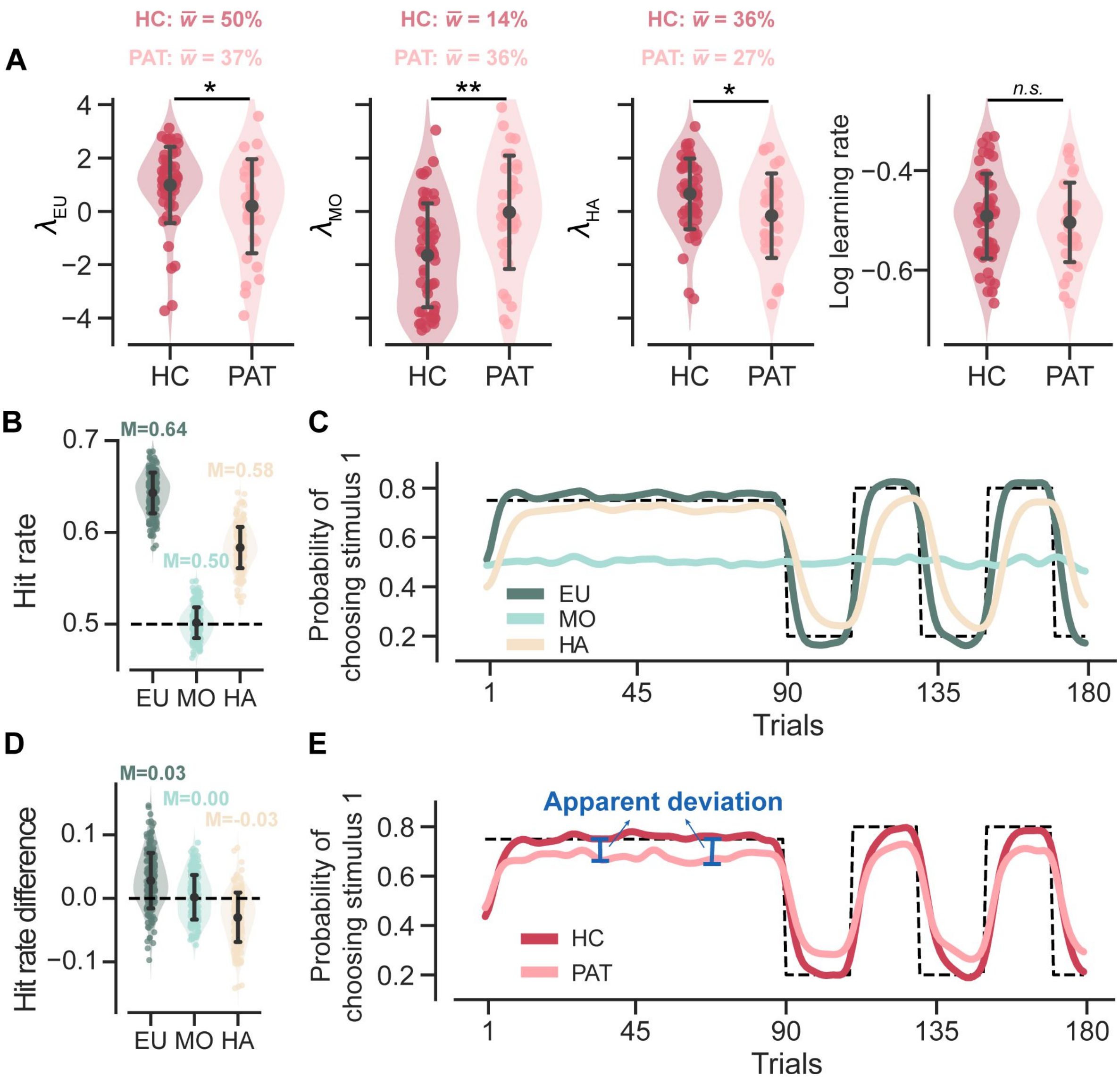

Figure 4.

Parameter analyses of the MOS6 model and simulated behaviors for all three strategies. Significance symbol conventions are: *: $p < 0.05$; **: $p < 0.01$; ***: $p < 0.001$; *n.s.*: non-significant. Abbreviations: HC-healthy controls, PAT-patients.

A. The fitted weighting parameters and learning rate of the MOS6 model. The y-axis means averaged preference over different volatile contexts (volatile/stable) and feedback contexts (reward/aversive). w indicates the averaged weighting parameters for each participant group. Error bars denote the standard deviation of the estimated mean across 54 healthy controls and 32 patients, respectively.

B. Simulated hit rates for the three decision strategies. Error bars represent the standard deviation across 200 simulations. The 200 simulations were evenly divided between groups using parameters similar to the healthy control group and the patient group. The groups differed only in their strategy preference (differences in w_{EU}, w_{MO}, w_{HA}) while all other parameters remained constant. For more simulation details, refer to ***Method: Simulation details***.

C. Averaged simulated learning curve for each strategy across 200 simulations, smoothed with a Gaussian kernel (standard deviation of 2 trials).

D. Simulated hit rate differences between volatile and stable for the three decision strategies. Error bars represent the standard deviation across 200 simulations.

E. Simulated learning curves for the healthy controls and patients, each averaged from 100 simulations within the group and smoothed with a Gaussian kernel (standard deviation of 2 trials).

($F(1, 77) = 0.393$, $p = 0.533$, $\eta^2 = 0.003$). Most importantly, the MOS22 model revealed no learning rate adaptation effect, as indicated by the learning rate parameters in the volatile context not being significantly larger than that in the stable context ($F(1, 77) = 0.126$, $p = 0.724$, $\eta^2 < 0.001$). Based on these findings, we drew two conclusions. First, MOS6 and MOS22 made consistent descriptions of participants' strategy preference during decision: the behavioral differences between two participant groups were mainly attributed to differences in their strategy preferences, rather than their learning rates. Second, the learning rate adaptation effect may be simply explained by context-free strategy preferences. We will further explain this second point in later sections. All statistical results about MOS22 are reported in ***Supplemental Note 4***.

Understanding patients' inferior task performances through strategy preferences

In this section, we illustrate how strategy preferences account for the three learning behavioral differences observed between the two participant groups, as shown in **Fig. 2**. To better understand how each decision strategy influences the three behavioral patterns, we simulated the MOS6 model using the median fitted parameters and outputted the decisions of each strategy (see simulation details in ***Methods***).

For hit rate, our simulations showed that the EU strategy achieved the highest hit rate, while the MO strategy basically performed at the chancel-level (**Fig. 4B**). These results are intuitively understandable. Since hit rate is defined based on feedback probability, the EU strategy, which actively tracks this probability, should be able to achieve a high hit rate. In contrast, the MO strategy, which completely ignores feedback probability, should achieve a chance-level hit rate. Interestingly, our simulations also showed that the HA strategy achieved an above-chance hit rate. This is because, although the HA strategy appears not to consider feedback probability directly, it still somewhat tracks feedback probability by simply repeating the past choices made by the EU. Accordingly, assigning lower weights to the two higher-hit-rate strategies, EU and HA (i.e., higher weighting on MO), naturally leads to inferior performance in the patients (**Fig. 2A**).

We also visualized the simulated learning curves for each strategy (averaged across the two groups) throughout the task (**Fig. 4C**). In both stable and volatile contexts, the EU strategy quickly approximates and converges to the true feedback probability. The HA strategy takes more trials to approach the true feedback probability, exhibiting slower learning. The MO strategy does not respond to environmental feedback, resulting in an almost flat learning curve. When the learning curves are combined separately for the two groups, we recover the seemingly slower learning curve in the patient group due to their stronger preference for the MO strategy (**Fig. 4E**). We also noted the larger apparent deviation from the true feedback probability in the patient group. These two features in **Fig. 2B** can thus be readily explained by the patients' stronger preference for the MO strategy, as the MO strategy does not learn feedback probability at all and exhibits a flat learning curve.

For the hit rate differences between the stable and volatile contexts, our simulations showed that the EU strategy achieves a higher hit rate in the volatile context than in the stable context (i.e., positive hit rate difference) (**Fig. 4D**). This is attributed to the EU strategy's active tracking of feedback probability (*i.e.*, the maximum possible hit rate), which increases from 75% in the stable context to 80% in the volatile context. Conversely, there were no changes in the MO strategy's hit rate from the stable to volatile contexts (i.e., 0 hit rate difference) because MO does not track feedback probability. Additionally, we found that the hit rate of the HA strategy was higher in the stable context than in the volatile (i.e., negative hit rate difference). This is possibly because the HA strategy requires more time to relearn true probability (**Fig. 4C**), particularly in the volatile context where the true probability frequently flips. Based on this, we can roughly estimate the hit rate difference for the healthy control group as ~ 0.042 $(\overline{w}_{\mathrm{EU}}^{\mathrm{HC}} \times 0.3 + \overline{w}_{\mathrm{MO}}^{\mathrm{HC}} \times 0 + \overline{w}_{\mathrm{HA}}^{\mathrm{HC}} \times$

$$-0.3 = 0.5 \times 0.3 + 0 + 0.36 \times -0.3)$$ and for the patient group as

$$\sim 0.030 \ (\overline{w}_{\mathrm{EU}}^{\mathrm{PAT}} \times 0.3 + \ \overline{w}_{\mathrm{MO}}^{\mathrm{PAT}} \times 0 + \ \overline{w}_{\mathrm{HA}}^{\mathrm{PAT}} \times -0.3 = 0.37 \times 0.3 + 0 + .27 \times -0.3 \ ).$$

This explains why healthy controls exhibited slightly a larger hit rate difference than the patient participants (**Fig. 2C**).

Atypical strategy preferences are connected to the general severity of anxiety and depression

We investigated the relationship between strategy preferences of the MOS6 model and symptom severity in the patient group (**Fig. 5**). Our findings indicate that patients with severe symptoms exhibit a weaker preference for the cognitively demanding EU strategy (Pearson's r = -0.221, p = 0.040) and a stronger preference for the simpler MO strategy (Pearson's r = 0.360, p = 0.001). Additionally, there was a significant correlation between symptom severity and the preference for the HA strategy (Pearson's r = -0.285, p = 0.007). These results highlight the strong clinical relevance of strategy preferences. See ***Supplemental Note 4*** for the same analysis conducted on the MOS22.

For completeness, we examined the correlation between learning rate adaptation (log volatile learning rate - log stable learning rate) and symptom severity within the MOS22 model. Not surprisingly, we found no significant correlation ($r(86)$ = 0.130, p = 0.233), which is consistent with our finding of no difference in learning rates across the two volatile contexts.

Strategy preferences may explain the learning rate differences across groups and contexts

Previous studies using probabilistic reversal learning tasks have made three major conclusions about learning rate. First, it has been documented that individuals with anxiety and depression have a smaller learning rate parameter (Chen et al., 2015; Pike & Robinson, 2022), thereby exhibiting a slower learning curve (**Fig. 2C**) and, possibly, a lower hit rate (**Fig. 2A**). Second, human participants have been found to be able to flexibly increase their learning rate in response to high environmental volatility (Behrens et al., 2007). Third, patient participants may involve with a deficit in such learning rate adaptation (Browning et al., 2015; Gagne et al., 2020), exhibiting a lesser extent of increase (**Fig. 2B**).

However, we recognize two limitations in this learning rate interpretation. Firstly, a higher learning rate does not necessarily improve the hit rate; it may lead to overreacting to feedback from stimuli with low probabilities. Second, and more importantly, a reduced learning rate merely prolongs the time needed to approach the true probability (Boyd & Vandenberghe, 2004) but cannot explain the apparent deviation from true probability observed in patient participants (**Fig. 2C**). In contrast, the account of mixture of strategies can naturally account for these two phenomena. As mentioned in the previous section, the mixture of EU and MO results in both a seemingly lower learning curve and a larger apparent deviation.

Here, we further demonstrate the possibility that the behavioral differences caused by a mixture of strategy may be interpreted as learning rate adaptation. We used the MOS6 to synthesize behavioral data for agents resembling healthy controls and patients by controlling all parameters except for the decision weights. Specifically, we set the weights of w_{EU} to 60%, w_{MO} to 15% and w_{HA} to 25% for the healthy control group, and the weights of w_{EU} to 15%, w_{MO} to 60% and w_{HA} to 25% for the patient group, with all other parameters fixed to the median values across all participants (see more details in ***Methods***). We simulated each group 20 times, and the simulated data reproduced the slower learning curve and the more apparent difference in the patient group (**Fig. 6A**). We fit the simulated data generated by MOS6 with the FLR22 model and found significant differences in the (log) learning rate between stable and volatile contexts (paired

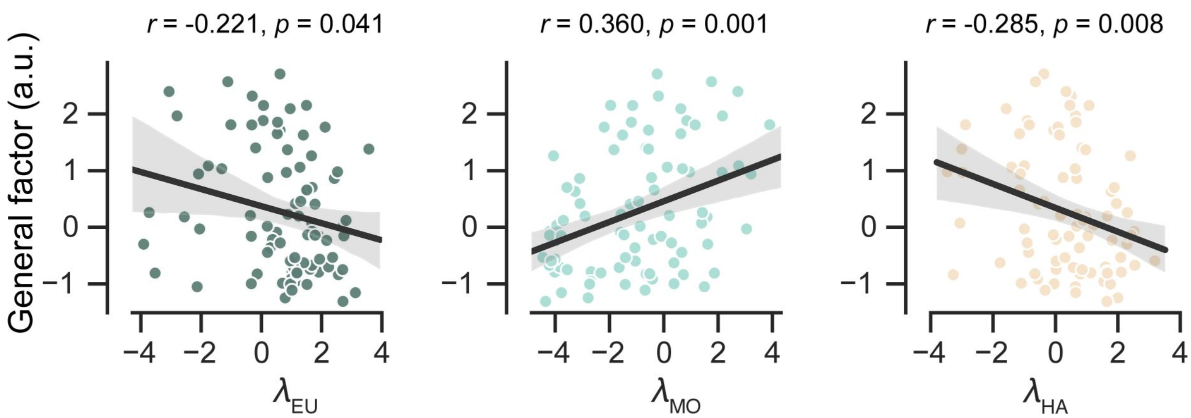

Figure 5.

Predict participants' symptom severity (*g* score) using strategy preferences of the MOS6 model. Each dot represents one participant. The shaded areas reflect 95% confidence intervals of the regression prediction.

t- test(39) = -3.217, $p = 0.003$, Cohen's d = 0.721; **Fig. 6B**). Furthermore, the agent resembling the patient group demonstrated a trend toward reduced learning rate adaptation compared to the agent resembling the healthy control group (**Fig. 6C**), consistent with the learning rate adaptation theory. These findings suggest that what might be perceived as learning rate adaptation could result from a mixture of strategy preferences. This observation also implies that strategy preferences may, at least partially, explain the maladaptive adaptations in learning rate observed in patients in response to environmental volatility.

Model and parameter recovery analyses support model and parameter identifiability in MOS

Although we have previously shown that the MOS models are quantitatively best-fitting models, there are two potential confounding factors. First, it is possible that differences in learning rate, rather than differences in strategy preference, could produce the same behavioral outcomes that are indistinguishable by the model fitting. If this holds, the MOS model might be problematic, as all learning rate differences may be automatically attributed to strategy preferences because of some unknown idiosyncratic model fitting mechanisms. Second, the fact that the MOS models outperform the others may be partly due to an unknown bias in the model design. It is possible that the MOS models always win, irrespective of how the data is generated.

To circumvent these issues, we conducted parameter recovery analyses on MOS6 and model recovery analyses on all models to investigate the identifiability of true parameters and models. For parameter recovery, we generated 80 synthetic datasets using 80 different parameter sets, each varying the four parameters of interest $\{\alpha_\psi, \lambda_{\mathrm{EU}}, \lambda_{\mathrm{MO}}, \lambda_{\mathrm{HA}}\}$, with the remaining parameters fixed to the median of the fitted parameters ($\beta = 10.803$, $\alpha_{\mathrm{HA}} = 0.423$). Each synthetic dataset consisted of 10 runs, resulting in a total of 800 synthetics runs (80 parameter sets × 10 runs). For each dataset, we fitted our MOS6 model and compared the fitted parameters to the ground-truth parameters. The parameter recovery results (**Fig. 7A**) demonstrate that the true parameters can be accurately estimated and identified (all Pearson's *r*s > 0.720), indicating that the effects of learning rate and weighting parameters are not interchangeable in the MOS6 model.

For model recovery, we sampled 40 participants (20 in each group) and used their fitted model parameters to generate synthetic datasets from each of the eight models. For each participant, we simulated 10 runs of behavioral data, resulting in a total of 3200 synthetic runs (8 generating models × 40 participants × 10 runs). We then fit all eight models to every dataset using the MAP method as the same before. The best-fitting model was always the one that generated the data, as indicated by all three quantitative metrics: AIC, BIC, and PXP (**Fig. 7B**). We note that the RS3 and PH4 models tend to account well for each other. The MOS6 model achieves good fitting performance on synthetic datasets generated by the RS3 and PH4 models, but not vice versa. The slight confusion between RS3, PH4, and MOS6 is because they all include the EU strategy. Most importantly, the MOS and FLR models cannot adequately account for each other's synthetic datasets, strongly supporting the independent computational effects of strategy preference and learning rate.

Discussion

In this article, we propose to understand humans' learning behaviors, especially the differences between healthy controls and patients, in the volatile reversal learning task through the lens of mixture-of-strategies. We develop the MOS model, which assumes that human participants make decisions by combining three distinct components: the EU, MO, and HA strategies. The EU strategy is rewarding but cognitively demanding, in contrast to the other two strategies, which are simpler and heuristic but less rewarding. Applying the MOS model to a public dataset, we show that the

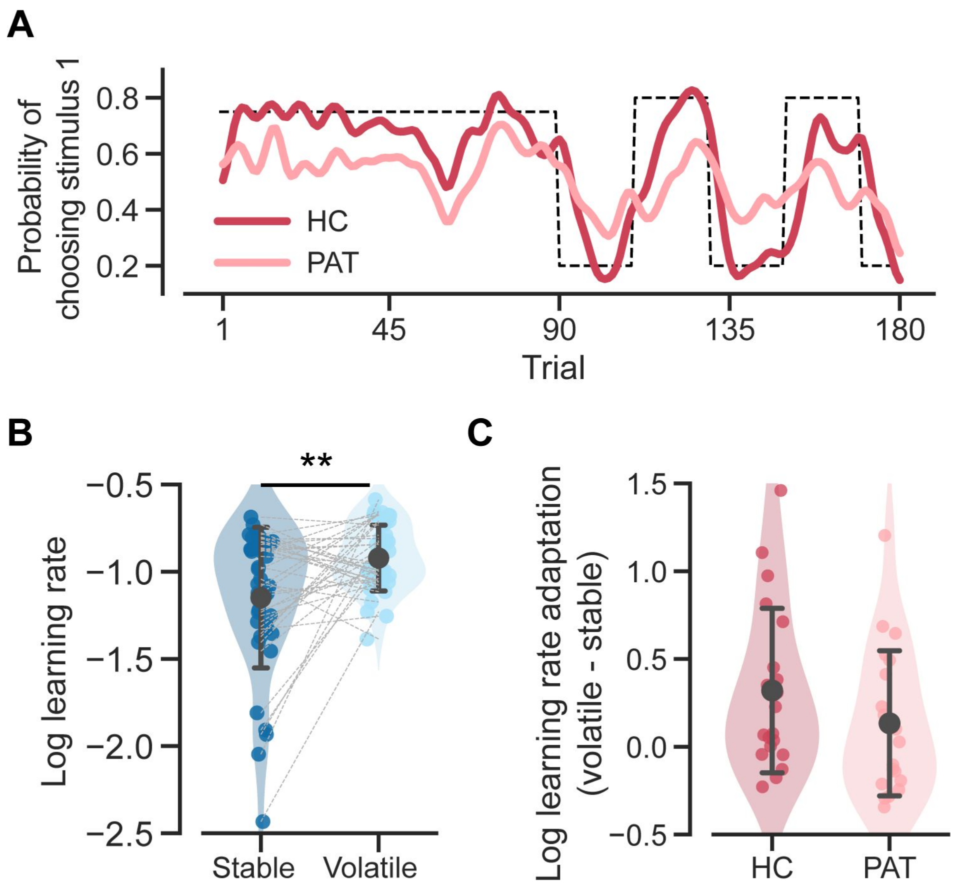

Figure 6.

Reproduction of the two learning rate adaptation effects using the MOS6 model. Significance symbol conventions are: *: *p* < 0.05; **: *p* < 0.01; HC represents the healthy-control-like agent; PAT represents the patient-like agent.

A. Simulated learning curves for the healthy controls and patients generated by the MOS6 model, smoothed with a Gaussian kernel (standard deviation of 2 trials).

B. The fitted FLR22 learning rate parameters for the stable context and volatile context. Error bars stand for the standard deviation across 40 synthesized datasets.

C. Learning rate adaptations, calculated by log volatile learning rate - log stable learning rate, for the healthy control-like agent and for the patient-like agent. Error bars stand for the standard deviation across 20 synthesized datasets.

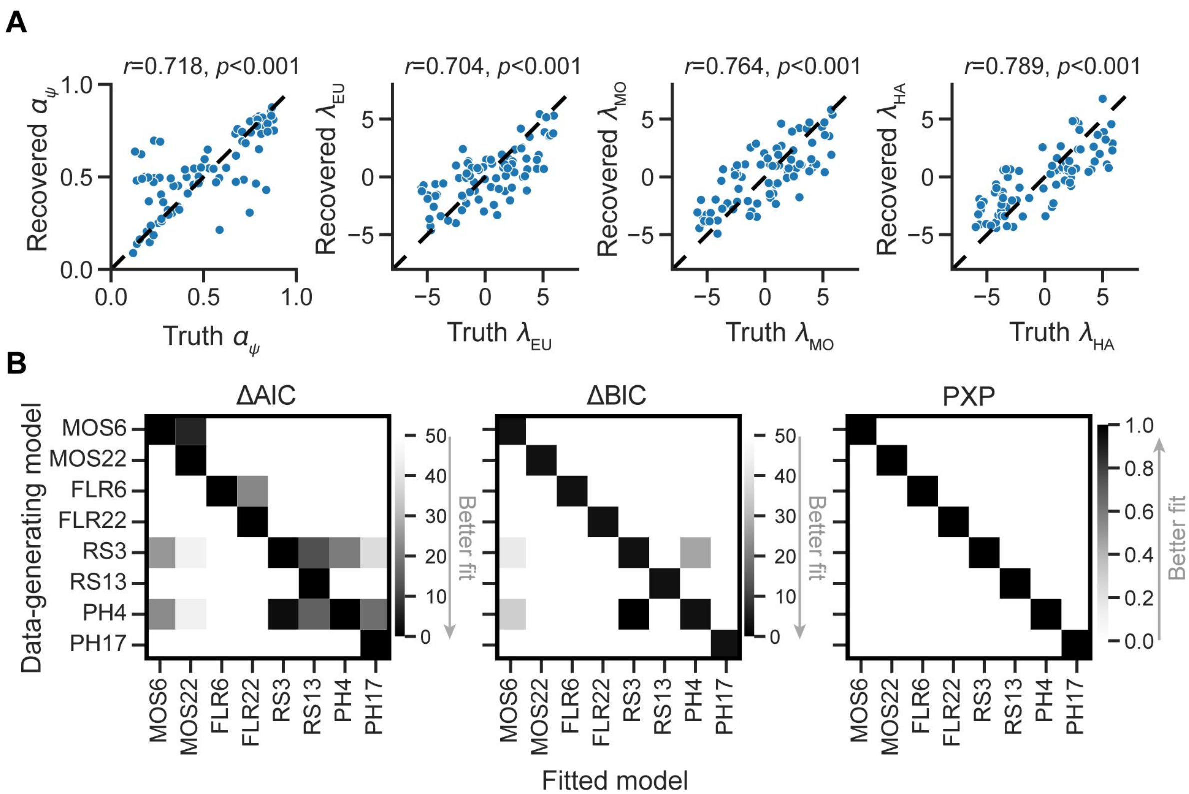

Figure 7.

Parameter and model recovery analyses.

A. Parameter recovery for the MOS6 model.

B. Model recovery analysis, showing the performance of models as evaluated by averaged relative AIC and BIC, as well as PXP scores for synthesized data generated from each of the 8 models. Darker tiles indicate better fits to the synthesized data.

MOS models can qualitatively capture several behavioral patterns that cannot be explained by previous models and quantitively better capture human behaviors and healthy-patient differences. The model reveals that individuals with MDD and GAD exhibit an atypical preference for simpler and less rewarding strategies (i.e., a stronger preference for the MO strategy), and this preference alone could explain their inferior task performance relative to healthy controls, as indicated by lower hit rates, reduced adaption volatility, and slower learning curves. Furthermore, we demonstrate the MOS model can reproduce the human behavioral learning rate adaptation effect without changing the learning rate itself. These findings suggest that a mixture of strategies provides an effective and parsimonious explanation for human learning behaviors in volatile reversal tasks.

The role of the HA strategy in volatile reversal learning

Although many observed behavioral differences can be explained by a shift in preference from the EU to the MO strategy among patients, we also explore the potential effects of the HA strategy. Compared to the MO, the HA strategy also saves cognitive resources but yields a significantly higher hit rate (**Fig. 4A**). Therefore, a preference for the HA over the MO strategy may reflect a more sophisticated balance between reward and complexity within an agent (Gershman, 2020): when healthier participants exhaust their cognitive resources for the EU strategy, they may cleverly resort to the HA strategy, adopting a simpler strategy but still achieving a certain level of hit rate. This explains the stronger preference for the HA strategy in the HC group (**Fig. 3A**) and the negative correlation between HA preferences and symptom severity (**Fig. 5**). Apart from shedding light on the cognitive impairments of patients, the inclusion of the HA strategy significantly enhances the model's fit to human behavior (see examples in Daw et al. (2011); Gershman (2020); and also ***Supplemental Note 1***).

Disassociate the learning rate adaptation and mixture of strategies

It is well-established that humans apply flexible learning rates in response to environmental volatility, exemplifying the successful application of ideal observer analysis. Behrens et al. (2007) constructed a hierarchical ideal Bayesian observer for the volatile reversal learning task that dynamically models how higher-order environmental volatility influences the updating speed of lower-order feedback probabilities. This model suggests that the human brain estimates environmental volatility and that humans are expected to exhibit a faster updating speed for feedback probabilities in volatile contexts. Consistent with their results, the context-dependent RS13 model revealed higher learning rate parameters in volatile contexts. Browning et al. (2015) identified the increase in learning rate from stable to volatile contexts as hallmark of human sensitivity to environmental volatility. They found that individuals with high trait anxiety showed reduced adaptations, thus indicating lower sensitivity to volatility. Gagne et al. (2020) extended this research to MDD and GAD patients receiving either reward or aversive feedback. Furthermore, the phenomenon of an increased learning rate from stable to volatile conditions has also been observed in other paradigms, such as the *Predictive Inference task* (Nassar et al., 2016; Nassar et al., 2010), where participants explicitly report their estimation of environmental statistics, allowing for a direct estimation of the learning rate.

Based on our findings, we applied the MOS model—an alternative but sufficiently accurate model—to the data collected from volatile reversal task and found that the expected learning rate adaptation was not observed. Instead, the MOS model points to an alternative explanation that accounts for multiple human behavioral patterns and their symptom severity in a more parsimonious manner, involving fewer parameters. More importantly, it is possible for the MOS model to capture the pattern of learning rate adaptation without necessitating actual changes in learning rates across different contexts. These findings indicate that future studies may systematically compare the account of learning rate adaptation and mixture-of-strategies.

It is important to note that learning rate adaptations and strategy preferences could simultaneously influence behavior. However, accurately describing both mechanisms, particularly in terms of differences between populations, requires more refined behavioral paradigms. For example, it would be helpful to use paradigms like the predictive inference task (Nassar et al., 2016; Nassar et al., 2010), which allows participants to directly report their learning rates, to minimize the confounding factors from the decision process.

Atypical learning speed in psychiatric diseases

In the present work, we found that individuals with depression and anxiety display apparent flatter learning curves in the probabilistic learning tasks (shown in **Fig. 4A**). We attributed this observation to participants' strategy preferences. However, in conventional Rescorla-Wagner modeling, learning speed is primarily indicated by the parameter of the learning rate. For example, Chen (2015) conducted a systematic review of reinforcement learning in patients with depression and identified 10 out of 11 behavioral datasets showing either comparable or slower learning rates in depressive patients. Nonetheless, depressive patients may not always exhibit slower learning rates. In a recent meta-analysis summarizing 27 articles with 3085 participants, including 1242 with depression and/or anxiety, Pike and Robinson (2022) found a reduced reward but enhanced aversive learning rate. This finding yields two practical implications. First, the heterogeneous findings in the literature may arise from heterogeneous pathologies in depression and anxiety. Second, some behavioral variations introduced by strategy preferences might have been misidentified as learning rate effects. The MOS model may provide useful complementary explanations for the consequences of a spectrum of symptoms.

Limitations and future directions

The MOS model is developed to offer context-free interpretations for the learning rate differences observed both between stable and volatile contexts and between healthy individuals and patients. However, we also recognize that the MOS account may not justify other learning rate effects based solely on strategy preferences. One such example is the valence-specific learning rate differences, where learning rates for better-than-expected outcomes are higher than those for worse-than-expected outcomes (Gagne et al., 2020). When fitted to the behavioral data, the context-dependent MOS22 model does not reveal valence-specific learning rates (***Supplemental Note 4***). Moreover, the valence-specific effect was not replicated in the FLR22 model when fitted to the synthesized data of MOS6.

The context-dependent MOS22 model revealed several weak interaction effects, suggesting an interaction between learning adaption and strategy preferences. For example, patients with MDD and GAD may find it too taxing to increase their learning rates in the volatile context, and instead resort to simpler strategy such as like MO as a compromise. Investigating this hypothesis may require a paradigm incorporating a self-reporting learning rate module, like the predictive inference task (Nassar et al., 2010), in volatile reversal learning tasks.

Theories suggest that humans increase learning rates in the volatile context due to increased perceived uncertainty about environmental statistics (Behrens et al., 2007; Nassar et al., 2010), while others propose that strategies enabling more exploration are preferred when managing uncertainty (Fan et al., 2023; Wilson et al., 2014). To explore these ideas, we may need to adjust the paradigm to offer a wider choice of stimuli, from two to three or four (*i.e.*, set size effects). Another question is why individuals with depression and anxiety tend toward simpler decisionmaking strategies. *Rumination*, a maladaptive emotion regulation behavior characterized by persistent negative thoughts observed in individuals with depression (Song et al., 2022; Yan et al., 2022), may consume cognitive resources, hindering the use of the more complex but rewarding EU strategy.

Conflict of Interests

The authors declare no competing financial interests.

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Author Contributions

Z. F. and R-Y.Z. conceived and designed the study. Z. F., M. Z., T.X., Y.L., H.X., and P.Q. processed the data. Z. F. implemented the computational models. Z. F., and R-Y.Z. made the first draft of the manuscript. All authors provide valuable feedback on the final manuscript.

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Editors

Reviewing Editor
Mimi Liljeholm
University of California, Irvine, Irvine, United States of America

Senior Editor
Michael Frank
Brown University, Providence, United States of America

Reviewer #2 (Public Review):

Summary:

Previous research shows that humans tend to adjust learning in environments where stimulus-outcome contingencies become more volatile. This learning rate adaptation is impaired in some psychiatric disorders, such as depression and anxiety. In this tudy the authors reanalyze previously published data on a reversal learning task with two volatility levels. Through a new model they provide some evidence for an alternative explanation whereby the learning rate adaptation is driven by different decision-making strategies and not learning deficits. In particular, they propose that adjusting of learning can be explained by deviations from the optimal decision-making strategy (based on maximizing expected utility) due to response stickiness or focus on reward magnitude. Furthermore, a factor related to general psychopathology of individuals with anxiety and depression negatively correlated with the weight on the optimal strategy and response stickiness, while it correlated positively with the magnitude strategy (a strategy that ignores the probability of outcome).

The main strength of the study is a novel and interesting explanation of an otherwise well-established finding in human reinforcement learning. This proposal is supported by rigorously conducted parameter retrieval and the comparison of the novel model to a wide range of previously published models. The authors explore from many angles, if and why the predictions from the new proposed model are superior to previously applied models.

My previous concerns were addressed in the revised version of the manuscript. I believe that the article now provides a new perspective on a well-established learning effect and offer a novel set of interesting response models that can be applied to a wide array of decision-making problems.

I see two limitations of the study not mentioned in the discussion of the manuscript. First, the task features binary inputs and responses, therefore unexpected uncertainty (volatility) is impossible to differentiate from the uncertainty about outcomes, and exploration is inseparable from random choices. Future work could validate these findings in task designs that allow to distinguish these processes. Second, clinical results are based on a small sample of patients and should be interpreted with this in mind.

https://doi.org/10.7554/eLife.93887.2.sa2

Reviewer #3 (Public Review):

Summary:

This paper presents a new formulation of a computational model of adaptive learning amid environmental volatility. Using a behavioral paradigm and data set made available by the authors of an earlier publication (Gagne et al., 2020), the new model is found to fit the data well. The model's structure consists of three weighted controllers that influence decisions on the basis of (1) expected utility, (2) potential outcome magnitude, and (3) habit. The model offers an interpretation of psychopathology-related individual differences in decision-making behavior in terms of differences in the relative weighting of the three controllers.

Strengths:

The newly proposed "mixture of strategies" (MOS) model is evaluated relative to the model presented in the original paper by Gagne et al., 2020 (here called the "flexible learning rate" or FLR model) and two other models. Appropriate and sophisticated methods are used for developing, parameterizing, fitting, and assessing the MOS model, and the MOS model performs well on multiple goodness-of-fit indices. Parameters of the model show decent recoverability and offer a novel interpretation for psychopathology-related individual differences. Most remarkably, the model seems to be able to account for apparent differences in behavioral learning rates between high-volatility and low-volatility conditions even with no true condition-dependent change in the parameters of its learning/decision processes. This finding calls into question a class of existing models that attribute behavioral adaptation to adaptive learning rates.

Weaknesses:

The authors have responded to the weaknesses noted previously.

https://doi.org/10.7554/eLife.93887.2.sa1

Author response:

The following is the authors' response to the original reviews.

> ***Public Reviews:***
>
> ***Reviewer #1:***
>
> *Point 1.1*
>
> *Summary: This paper describes a reanalysis of data collected by Gagne et al. (2020), who investigated how human choice behaviour differs in response to changes in environmental volatility. Several studies to date have demonstrated that individuals appear to increase their learning rate in response to greater volatility and that this adjustment is reduced amongst individuals with anxiety and depression. The present authors challenge this view and instead describe a novel Mixture of Strategies (MOS) model, that attributes individual differences in choice behaviour to different weightings of three distinct decision-making strategies. They demonstrate that the MOS model provides a superior fit to the data and that the previously observed differences between patients and healthy controls may be explained by patients opting for a less cognitively demanding, but suboptimal, strategy.*
>
> *Strengths:*
>
> *The authors compare several models (including the original winning model in Gagne et al., 2020) that could feasibly fit the data. These are clearly described and are evaluated using a range of model diagnostics. The proposed MOS model appears to provide a superior fit across several tests.*
>
> *The MOS model output is easy to interpret and has good face validity. This allows for the generation of clear, testable, hypotheses, and the authors have suggested several lines of*

> *potential research based on this.*

We appreciate the efforts in understanding our manuscript. This is a good summary.

> *Point 1.2*
>
> *The authors justify this reanalysis by arguing that learning rate adjustment (which has previously been used to explain choice behaviour on volatility tasks) is likely to be too computationally expensive and therefore unfeasible. It is unclear how to determine how "expensive" learning rate adjustment is, and how this compares to the proposed MOS model (which also includes learning rate parameters), which combines estimates across three distinct decision-making strategies.*

We are sorry for this confusion. Actually, our motivation is that previous models only consider the possibility of learning rate adaptation to different levels of environmental volatility. The drawback of previous computational modeling is that they require a large number of parameters in multi-context experiments. We feel that learning rate adaptation may not be the only mechanisms or at least there may exist alternative explanations. Understanding the true mechanisms is particularly important for rehabilitation purposes especially in our case of anxiety and depression. To clarify, we have removed all claims about the learning rate adaptation is “too complex to understand”.

> *Point 1.3*
>
> *As highlighted by the authors, the model is limited in its explanation of previously observed learning differences based on outcome value. It's currently unclear why there would be a change in learning across positive/negative outcome contexts, based on strategy choice alone.*

Thanks for mentioning this limitation. We want to highlight two aspect of work.

First, we developed the MOS6 model primarily to account for the learning rate differences between stable and volatile contexts, and between healthy controls and patients, not for between positive and negative outcomes. In the other words, our model does not eliminate the possibility of different learning rate in positive and negative outcomes.

Second, Figure 3A shows that FLR (containing different learning parameters for positive/negative outcomes) even performed worse than MOS6 (setting identical learning rate for positive/negative outcomes). This result question whether learning rate differences between positive/negative outcomes exist in our dataset.

Action: We now include this limitation in lines 784-793 in discussion:

“The MOS model is developed to offer context-free interpretations for the learning rate differences observed both between stable and volatile contexts and between healthy individuals and patients. However, we also recognize that the MOS account may not justify other learning rate effects based solely on strategy preferences. One such example is the valence-specific learning rate differences, where learning rates for better-than-expected outcomes are higher than those for worse-than-expected outcomes (Gagne et al., 2020). When fitted to the behavioral data, the context-dependent MOS22 model does not reveal valence-specific learning rates (Supplemental Note 4). Moreover, the valence-specific effect was not replicated in the FLR22 model when fitted to the synthesized data of MOS6.”

> *Point 1.4*

> *Overall the methods are clearly presented and easy to follow, but lack clarity regarding some key features of the reversal learning task.*
>
> *Throughout the method the stimuli are referred to as "right" and "left". It's not uncommon in reversal learning tasks for the stimuli to change sides on a trial-by-trial basis or counterbalanced across stable/volatile blocks and participants. It is not stated in the methods whether the shapes were indeed kept on the same side throughout. If this is the case, please state it. If it was not (and the shapes did change sides throughout the task) this may have important implications for the interpretation of the results. In particular, the weighting of the habitual strategy (within the Mixture of Strategies model) could be very noisy, as participants could potentially have been habitual in choosing the same side (i.e., performing the same motor movement), or in choosing the same shape. Does the MOS model account for this?*

We are sorry for the confusion. Yes, two shapes indeed changed sides throughout the task. We replaced the "left" and "right" with "stimulus 1" and "stimulus 2". We also acknowledge the possibility that participants may develop a habitual preference for a particular side, rather than a shape. Due to the counterbalance design, habitual on side will introduce a random selection noise in choices, which should be captured by the MOS model through the inverse temperature parameter.

> *Point 1.5*
>
> *Line 164: "Participants received points or money in the reward condition and an electric shock in the punishment condition." What determined whether participants received points or money, and did this differ across participants?*

Thanks! We have the design clarified in lines 187-188:

"Each participant was instructed to complete two blocks of the volatile reversal learning task, one in the reward context and the other in the aversive context",

and in lines:

"A total of 79 participants completed tasks in both feedback contexts. Four participants only completed the task in the reward context, while three participants only completed the aversive task."

> *Point 1.6*
>
> *Line 167: "The participant received feedback only after choosing the correct stimulus and received nothing else" Is this correct? In Figure 1a it appears the participant receives feedback irrespective of the stimulus they chose, by either being shown the amount 1-99 they are being rewarded/shocked, or 0. Additionally, what does the "correct stimulus" refer to across the two feedback conditions? It seems intuitive that in the reward version, the correct answer would be the rewarding stimulus - in the loss version is the "correct" answer the one where they are not receiving a shock?*

Thanks for raising this issue. We removed the term "correct stimulus" and revised the lines 162-166 accordingly:

"Only one of the two stimuli was associated with actual feedback (0 for the other one). The feedback magnitude, ranged between 1-99, is sampled uniformly and independently for each shape from trial to trial. Actual feedback was delivered only if the stimulus associated with feedback was chosen; otherwise, a number "0" was displayed on the screen, signifying that the chosen stimulus returns nothing."

> *Point 1.7*
>
> *Line 176: "The whole experiment included two runs each for the two feedback conditions." Does this mean participants completed the stable and volatile blocks twice, for each feedback condition? (i.e., 8 blocks total, 4 per feedback condition).*

Thanks! We have removed the term “block”, and now we refer to it as “context”. In particular, we removed phrases like “stable block” and “volatile block” and used “context” instead.

Action: See lines 187-189 for the revised version.

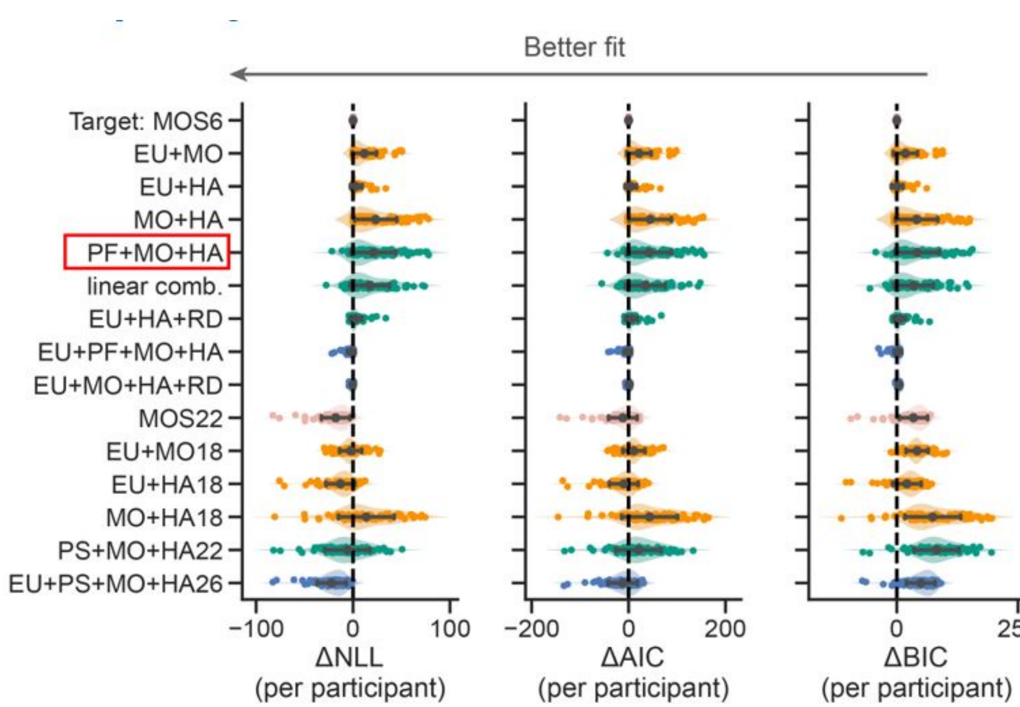
“Each participant was instructed to complete two runs of the volatile reversal learning task, one in the reward context and the other in the aversive context. Each run consisted of 180 trials, with 90 trials in the stable context and 90 in the volatile context (Fig. 1B).”

> *Point 1.8*
>
> *In the expected utility (EU) strategy of the Mixture or Strategies model, the expected value of the stimulus on each trial is produced by multiplying the magnitude and probability of reward/shock. In Gagne et al.'s original paper, they found that an additive mixture of these components better-captured participant choice behaviour - why did the authors not opt for the same strategy here?*

Thanks for asking this. Their strategy basic means the mixture of PF+MO+HA, where PF stands for the feedback probability (e.g., 0.3 or 0.7) without multiplying feedback magnitude. However, ours are EU+MO+HA, where EU stands for feedback probability x feedback magnitude. We did compare these two strategies and the model using their strategy performed much worse than ours (see the red box below).

Author response image 1.

Thorough model comparison.



> *Point 1.9*
>
> *How did the authors account for individuals with poor/inattentive responding, my concern is that the habitual strategy may be capturing participants who did not adhere to the task (or is this impossible to differentiate?).*

The current MOS6 model distinguishes between the HA strategy and the inattentive response. Due to the counterbalance design, the HA strategy requires participants to actively track the stimuli on the screen. In contrast, the inattentive responding, like the same motor movement mentioned in Point 1.4, should exhibit random selection in their behavioral data, which should be account by the inverse temperature parameter.

> *Point 1.10*
>
> *The authors provide a clear rationale for, and description of, each of the computational models used to capture participant choice behaviour.*
>
> *• Did the authors compare different combinations of strategies within the MOS model (e.g., only including one or two strategies at a time, and comparing fit?) I think more explanation is needed as to why the authors opted for those three specific strategies.*

We appreciate this great advice. Following your advice, we conducted a thorough model comparisons. Please refer to Figure R1 above. The detailed text descriptions of all the models in Figure R1 are included in Supplemental Note 1.

> *Point 1.11*
>
> *Please report the mean and variability of each of the strategy weights, per group.*

Thanks. We updated the mean of variability of the strategies in lines 490-503:

"We first focused on the fitted parameters of the MOS6 model. We compared the weight parameters (, ,) across groups and conducted statistical tests on their logits (, ,). The patient group showed a ~37% preference towards the EU strategy, which is significantly weaker than the ~50% preference in healthy controls (healthy controls' : M = 0.991, SD = 1.416; patients' : M = 0.196, SD = 1.736; t(54.948) = 2.162, p = 0.035, Cohen's d = 0.509; Fig. 4A). Meanwhile, the patients exhibited a weaker preference (~27%) for the HA strategy compared to healthy controls (~36%) (healthy controls' : M = 0.657, SD = 1.313; patients' : M = -0.162, SD = 1.561; t(56.311) = 2.455, p = 0.017, Cohen's d = 0.574), but a stronger preference for the MO strategy (36% vs. 14%; healthy controls' : M = -1.647, SD = 1.930; patients' : M = -0.034, SD = 2.091; t(63.746) = -3.510, p = 0.001, Cohen's d = 0.801). Most importantly, we also examined the learning rate parameter in the MOS6 but found no group differences (t(68.692) = 0.690, p = 0.493, Cohen's d = 0.151). These results strongly suggest that the differences in decision strategy preferences can account for the learning behaviors in the two groups without necessitating any differences in learning rate per se."

> *Point 1.12*
>
> *The authors compare the strategy weights of patients and controls and conclude that patients favour more simpler strategies (see Line 417), based on the fact that they had higher weights for the MO, and lower on the EU.*
>
> *(1) However, the finding that control participants were more likely to use the habitual strategy was largely ignored. Within the control group, were the participants significantly more likely to opt for the EU strategy, over the HA? 2) Further, on line 467 the authors*

> *state "Additionally, there was a significant correlation between symptom severity and the preference for the HA strategy (Pearson's r = -0.285, p = 0.007)." Apologies if I'm mistaken, but does this negative correlation not mean that the greater the symptoms, the less likely they were to use the habitual strategy?*
>
> *I think more nuance is needed in the interpretation of these results, particularly in the discussion.*

Thanks. The healthy participants seemed more likely to opt for the EU strategy, although this difference did not reach significance (paired-t(53) = 1.258, $p = 0.214$, Cohen's d = 0.242). We systematically explore the role of HA. Compared to the MO, the HA saves cognitive resources but yields a significantly higher hit rate (Fig. 4A). Therefore, a preference for the HA over the MO strategy may reflect a more sophisticated balance between reward and complexity within an agent: when healthier subjects run out of cognitive resources for the EU strategy, they will cleverly resort to the HA strategy, adopting a simpler strategy but still achieving a certain level of hit rate. This explains the negative symptom-HA correlation. As clever as the HA strategy is, it is not surprising that the health control participants opt more for the HA during decision-making.

However, we are cautious to draw strong conclusion on (1) non-significant difference between EU and HA within health controls and (2) the negative symptom-HA correlation. The reason is that the MOS22, the context-dependent variant, 1) exhibited a significant higher preference for EU over HA (paired-t(53) = 4.070, $p < 0.001$, Cohen's d = 0.825) and 2) did not replicate this negative correlation (Supplemental Information Figure S3).

Action: Simulation analysis on the effects of HA was introduced in lines 556-595 and Figure 4. We discussed the effects of HA in lines 721-733:

"Although many observed behavioral differences can be explained by a shift in preference from the EU to the MO strategy among patients, we also explore the potential effects of the HA strategy. Compared to the MO, the HA strategy also saves cognitive resources but yields a significantly higher hit rate (Fig. 4A). Therefore, a preference for the HA over the MO strategy may reflect a more sophisticated balance between reward and complexity within an agent (Gershman, 2020): when healthier participants exhaust their cognitive resources for the EU strategy, they may cleverly resort to the HA strategy, adopting a simpler strategy but still achieving a certain level of hit rate. This explains the stronger preference for the HA strategy in the HC group (Fig. 3A) and the negative correlation between HA preferences and symptom severity (Fig. 5). Apart from shedding light on the cognitive impairments of patients, the inclusion of the HA strategy significantly enhances the model's fit to human behavior (see examples in Daw et al. (2011); Gershman (2020); and also Supplemental Note 1 and Supplemental Figure S3)."

> *Point 1.13*
>
> *Line 513: "their preference for the slowest decision strategy" - why is the MO considered the slowest strategy? Is it not the least cognitively demanding, and therefore, the quickest?*

Sorry for the confusion. In Fig. 5C, we conducted simulations to estimate the learning speed for each strategy. As shown below, the MO strategy exhibits a flat learning curve. Our claim on the learning speed was based solely on simulation outcomes without referring to cognitive demands. Note that our analysis did not aim to compare the cognitive demands of the MO and HA strategies directly.

Action: We explain the learning speed of the three strategies in lines 571-581.

> *Point 1.14*
>
> *The authors argue that participants chose suboptimal strategies, but do not actually report task performance. How does strategy choice relate to the performance on the task (in terms of number of rewards/shocks)? Did healthy controls actually perform any better than the patient group?*

Thanks for the suggestion. The answers are: 1) EU is the most rewarding > the HA > the MO (Fig. 5A), and 2) yes healthy controls did actually perform better than patients in terms of hit rate (Fig. 2).

Action: We included additional sections on above analyses in lines 561-570 and lines 397-401.

> *Point 1.15*
>
> *The authors speculate that Gagne et al. (2020) did not study the relationship between the decision process and anxiety and depression, because it was too complex to analyse. It's unclear why the FLR model would be too complex to analyse. My understanding is that the focus of Gagne's paper was on learning rate (rather than noise or risk preference) due to this being the main previous finding.*

Thanks! Yes, our previous arguments are vague and confusing. We have removed all this kind of arguments.

> *Point 1.16*
>
> *Minor Comments:*
>
> *• Line 392: Modeling fitting > Model fitting*
>
> *• Line 580 reads "The MO and HA are simpler heuristic strategies that are cognitively demanding."*
>
> *- should this read as less cognitively demanding?*
>
> *• Line 517: health > healthy*
>
> *• Line 816: Desnity > density*

Sorry for the typo! They have all been fixed.

> ***Reviewer #2:***
>
> *Point 2.1*
>
> *Summary: Previous research shows that humans tend to adjust learning in environments where stimulus-outcome contingencies become more volatile. This learning rate adaptation is impaired in some psychiatric disorders, such as depression and anxiety. In this study, the authors reanalyze previously published data on a reversal-learning task with two volatility levels. Through a new model, they provide some evidence for an alternative explanation whereby the learning rate adaptation is driven by different decision-making strategies and not learning deficits. In particular, they propose that adjusting learning can be explained by deviations from the optimal decision-making strategy (based on maximizing expected utility) due to response stickiness or focus on reward magnitude. Furthermore, a factor related to the general psychopathology of individuals with anxiety and depression negatively correlated with the weight on the*

> *optimal strategy and response stickiness, while it correlated positively with the magnitude strategy (a strategy that ignores the probability of outcome).*

Thanks for evaluating our paper. This is a good summary.

> *Point 2.2*
>
> *My main concern is that the winning model (MOS6) does not have an error term (inverse temperature parameter beta is fixed to 8.804).*
>
> *(1) It is not clear why the beta is not estimated and how were the values presented here chosen. It is reported as being an average value but it is not clear from which parameter estimation. Furthermore, with an average value for participants that would have lower values of inverse temperature (more stochastic behaviour) the model is likely overfitting.*
>
> *(2) In the absence of a noise parameter, the model will have to classify behaviour that is not explained by the optimal strategy (where participants simply did not pay attention or were not motivated) as being due to one of the other two strategies.*

We apologize for any confusion caused by our writing. We did set the inverse temperature as a free parameter and quantitatively estimate it during the model fitting and comparison. We also created a table to show the free parameters for each models. In the previous manuscript, we did mention “temperature parameter beta is fixed to 8.804”, but only for the model simulation part, which is conducted to interpret some model behaviors.

We agree with the concern that using the averaged value over the inverse temperature could lead to overfitting to more stochastic behaviors. To mitigate this issue, we now used the median as a more representative value for the population during simulation. Nonetheless, this change does not affect our conclusion (see simulation results in Figures 4&6).

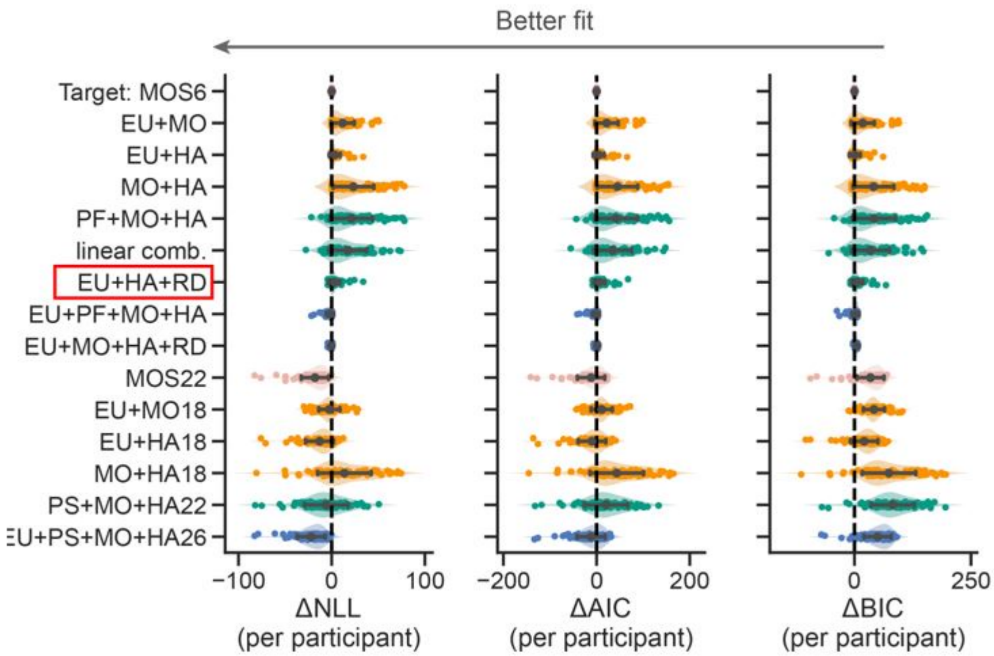
Action: We now use the term “free parameter” to emphasize that the inverse temperature was fitted rather than fixed. We also create a new table “Table 1” in line 458 to show all the free parameters within a model. We also update the simulation details in lines 363-391 for more clarifications.

> *Point 2.3*
>
> *(3) A model comparison among models with inverse temperature and variable subsets of the three strategies (EU + MO, EU + HA) would be interesting to see. Similarly, comparison of the MOS6 model to other models where the inverse temperature parameter is fixed to 8.804).*
>
> *This is an important limitation because the same simulation as with the MOS model in Figure 3b can be achieved by a more parsimonious (but less interesting) manipulation of the inverse temperature parameter.*

Thanks, we added a comparison between the MOS6 and the two lesion models (EU + MO, EU + HA). Please refer to the figure below and Point 1.8.

We also realize that the MO strategy could exhibit averaged learning curves similar to random selection. To confirm that patients' slower learning rates are due to a preference for the MO strategy, we compared the MOS6 model with a variant (see the red box below) in which the MO strategy is replaced by Random (RD) selection that assigns a 0.5 probability to both choices. This comparison showed that the original MOS6 model with the MO strategy better fits human data.

Author response image 2.



> *Point 2.4*
>
> *Furthermore, the claim that the EU represents an optimal strategy is a bit overstated. The EU strategy is the only one of the three that assumes participants learn about the stimulus-outcomes contingencies. Higher EU strategy utilisation will include participants that are more optimal (in maximum utility maximisation terms), but also those that just learned better and completely ignored the reward magnitude.*

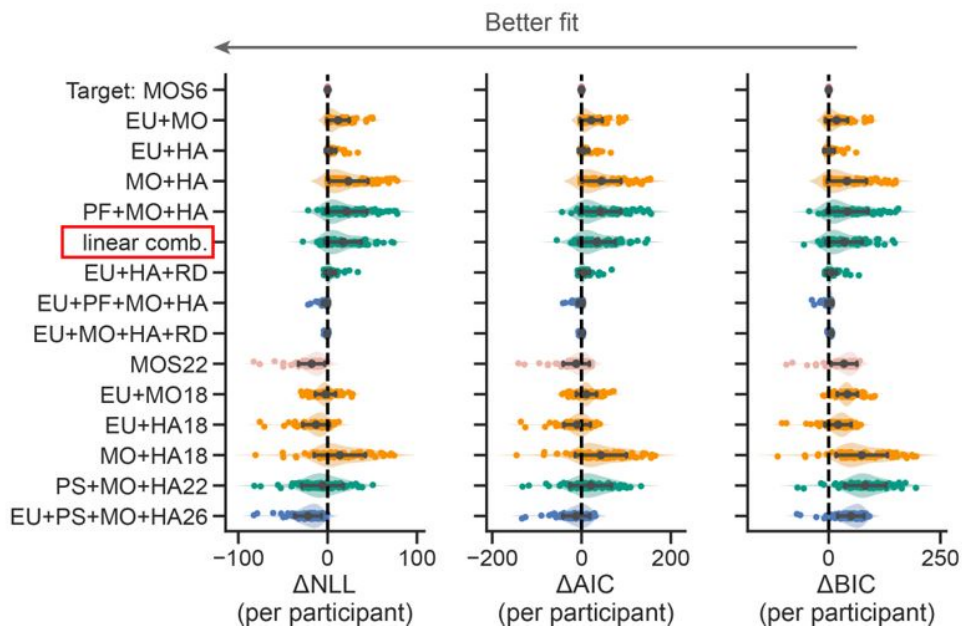
Thank you for your feedback. We have now revised the paper to remove all statement about “EU strategy is the optimal” and replaced by “EU strategy is rewarding but complex”. We agree that both the EU strategy and the strategy only focusing on feedback probability (i.e., ignoring the reward magnitude, refer to as the PF strategy) are rewarding but complex beyond two simple heuristics. We also included the later strategy in our model comparisons (see the next section Point 2.5).

> *Point 2.5*
>
> *The mixture strategies model is an interesting proposal, but seems to be a very convoluted way to ask: to what degree are decisions of subjects affected by reward, what they've learned, and response stickiness? It seems to me that the same set of questions could be addressed with a simpler model that would define choice decisions through a softmax with a linear combination of the difference in rewards, the difference in probabilities, and a stickiness parameter.*

Thanks for suggesting this model. We did include the proposed linear combination models (see “linear comb.” in the red box below) and found that it performed significantly worse than the MOS6.

Action: We justified our model selection criterion in the Supplemental Note 1.

Author response image 3.



> *Point 2.6*
>
> *Learning rate adaptation was also shown with tasks where decision-making strategies play a less important role, such as the Predictive Inference task (see for instance Nassar et al, 2010). When discussing the merit of the findings of this study on learning rate adaptation across volatility blocks, this work would be essential to mention.*

Thanks for mentioning this great experimental paradigm, which provides an ideal solution for disassociating the probability learning and decision process. We have discussed about this paradigm as well as the associated papers in discussion lines 749-751, 763-765, and 796-801.

> *Point 2.7*
>
> *Minor mistakes that I've noticed:*
>
> *Equation 6: The learning rate for response stickiness is sometimes defined as alpha_AH or alpha_pi.*
>
> *Supplementary material (SM) Contents are lacking in Note1. SM talks about model MOS18, but it is not defined in the text (I am assuming it is MOS22 that should be talked about here).*

Thanks! Fixed.

> ***Reviewer #3:***
>
> *Point 3.1*
>
> *Summary: This paper presents a new formulation of a computational model of adaptive learning amid environmental volatility. Using a behavioral paradigm and data set made available by the authors of an earlier publication (Gagne et al., 2020), the new model is found to fit the data well. The model's structure consists of three weighted controllers*

> *that influence decisions on the basis of (1) expected utility, (2) potential outcome magnitude, and (3) habit. The model offers an interpretation of psychopathology-related individual differences in decision-making behavior in terms of differences in the relative weighting of the three controllers.*
>
> *Strengths: The newly proposed "mixture of strategies" (MOS) model is evaluated relative to the model presented in the original paper by Gagne et al., 2020 (here called the "flexible learning rate" or FLR model) and two other models. Appropriate and sophisticated methods are used for developing, parameterizing, fitting, and assessing the MOS model, and the MOS model performs well on multiple goodness-of-fit indices. The parameters of the model show decent recoverability and offer a novel interpretation for psychopathology-related individual differences. Most remarkably, the model seems to be able to account for apparent differences in behavioral learning rates between high-volatility and low-volatility conditions even with no true condition-dependent change in the parameters of its learning/decision processes. This finding calls into question a class of existing models that attribute behavioral adaptation to adaptive learning rates.*

Thanks for evaluating our paper. This is a good summary.

> *Point 3.2*
> *(1) Some aspects of the paper, especially in the methods section, lacked clarity or seemed to assume context that had not been presented. I found it necessary to set the paper down and read Gagne et al., 2020 in order to understand it properly.*
>
> *(3) Clarification-related suggestions for the methods section:*
> *- Explain earlier that there are 4 contexts (reward/shock crossed with high/low volatility). Lines 252-307 contain a number of references to parameters being fit separately per context, but "context" was previously used only to refer to the two volatility levels.*

Action: We have placed the explanation as well as the table about the 4 contexts (stable-reward/stable-aversive/volatile-reward/volatile-aversive) earlier in the section that introduces the experiment paradigm (lines 177-186):

"Participants was supposed to complete this learning and decision-making task in four experimental contexts (Fig. 1A), two feedback contexts (reward or aversive) two volatility contexts (stable or volatile). Participants received points in the reward context and an electric shock in the aversive context. The reward points in the reward context were converted into a monetary bonus by the end of the task, ranging from £0 to £10. In the stable context, the dominant stimulus (i.e., a certain stimulus induces the feedback with a higher probability) provided a feedback with a fixed probability of 0.75, while the other one yielded a feedback with a probability of 0.25. In the volatile context, the dominant stimulus's feedback probability was 0.8, but the dominant stimulus switched between the two every 20 trials. Hence, this design required participants to actively learn and infer the changing stimulus-feedback contingency in the volatile context."

> *- It would be helpful to provide an initial outline of the four models that will be described since the FLR, RS, and PH models were not foreshadowed in the introduction. For the FLR model in particular, it would be helpful to give a narrative overview of the components of the model before presenting the notation.*

Action: We now include an overview paragraph in the section of computation model to outline the four models as well as the hypotheses constituted in the model (lines 202-220).

> *- The subsection on line 343, describing the simulations, lacks context. There are references to three effects being simulated (and to "the remaining two effects") but these*

> *are unclear because there's no statement in this section of what the three effects are.*
>
> *- Lines 352-353 give group-specific weighting parameters used for the stimulations of the HC and PAT groups in Figure 4B. A third, non-group-specific set of weighting parameters is given above on lines 348-349. What were those used for?*
>
> *- Line 352 seems to say Figure 4A is plotting a simulation, but the figure caption seems to say it is plotting empirical data.*

These paragraphs has been rewritten and the abovementioned issues have been clarified. See lines 363-392.

> *Point 3.2*
>
> *(2) There is little examination of why the MOS model does so well in terms of model fit indices. What features of the data is it doing a better job of capturing? One thing that makes this puzzling is that the MOS and FLR models seem to have most of the same qualitative components: the FLR model has parameters for additive weighting of magnitude relative to probability (akin to the MOS model's magnitude-only strategy weight) and for an autocorrelative choice kernel (akin to the MOS model's habit strategy weight). So it's not self-evident where the MOS model's advantage is coming from.*

An intuitive understanding of the FLR model is that it estimates the stimuli value through a linear combination of probability feedback (PF, $[\psi(s_1) - \psi(s_2)]$)and (non-linear) magnitude ($|m(s_1) - m(s_2)|^r$). .See equation:

$$v = \lambda[\psi(s_1) - \psi(s_2)] + (1 - \lambda)\text{sign}\big(m(s_1) - m(s_2)\big)|m(s_1) - m(s_2)|^r$$

Also, the FLR model include the mechanisms of HA as:

$$\pi(s_1|v, \pi_{\text{HA}}) = \frac{1}{1 + \exp(-\beta v - \beta_{\text{HA}}[\pi_{\text{HA}}(s_1) - \pi_{\text{HA}}(s_2)])}$$

In other words, FLR model considers the mechanisms about the probability of feedback (PF)+MO+HA (see Eq. XX in the original study), but our MOS considers the mechanisms of EU+MO+HA. The key qualitative difference lies between FLR and MOS is the usage of the expected utility formula (EU) instead the probability of feedback (PF). The advantage of our MOS model has been fully evidenced by our model comparisons, indicating that human participants multiply probability and magnitude rather than only considering probability. The EU strategy has also been suggested by a large pile of literature (Gershman et al., 2015; Von Neumann & Morgenstern, 1947).

Making decisions based on the multiplication of feedback probability and magnitude can often yield very different results compared to decisions based on a linear combination of the two, especially when the two magnitudes have a small absolute difference but a large ratio. Let’s consider two cases:

(1) Stimulus 1: $\psi_1 = 0.75, m_1 = 0.1$ vs. Stimulus 2: $\psi_1 = 0.25, m_1 = 0.4$

(2) Stimulus 1: $\psi_1 = 0.75, m_1 = 0.2$ vs. Stimulus 2: $\psi_1 = 0.25, m_1 = 0.8$

The EU strategy may opt for stimulus 2 in both cases, since stimulus 2 always has a larger expected value. However, it is very likely for the PF+MO to choose stimulus 1 in the first case.

For example, when $\lambda = 0.5$, $\lambda\psi_1 + (1-\lambda)m_1 = 0.425 > \lambda\psi_2 + (1-\lambda)m_2 = 0.325$. If we want the PF+MO to also choose stimulus to align with the EU strategy, we need to increase the weight on magnitude $(1-\lambda)$. Note that in this example we divided the magnitude value by 100 to ensure that probability and magnitude are on the same scale to help illustration.

In the dataset reported by Gagne, 2020, the described scenario seems to occur more often in the aversive context than in the reward context. To accurately capture human behaviors, FLR22 model requires a significantly larger weight for magnitude in the aversive context $(1-\lambda \approx 0.4)$ than in the reward context $(1-\lambda \approx 0.2)$. Interestingly, when the weights for magnitude in different contexts are forced to be equal, the model (FLR6) fails, exhibiting an almost chance-level performance throughout learning (Fig. 3E, G). In contrast, the MOS6 model, and even the RS3 model, exhibit good performance using one identical set of parameters across contexts. Both MOS6 and RS3 include the EU strategy during decision-making. These findings suggest humans make decisions using the EU strategy rather than PF+MO.

The focus of our paper is to present that a good-enough model can interpret the same dataset in a completely different perspective, not necessarily to explore improvements for the FLR model.

> *Point 3.3*
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> *One of the paper's potentially most noteworthy findings (Figure 5) is that when the FLR model is fit to synthetic data generated by the expected utility (EU) controller with a fixed learning rate, it recovers a spurious difference in learning rate between the volatile and stable environments. Although this is potentially a significant finding, its interpretation seems uncertain for several reasons:*
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> *- According to the relevant methods text, the result is based on a simulation of only 5 task blocks for each strategy. It would be better to repeat the simulation and recovery multiple times so that a confidence interval or error bar can be estimated and added to the figure.*
>
> *- It makes sense that learning rates recovered for the magnitude-oriented (MO) strategy are near zero, since behavior simulated by that strategy would have no reason to show any evidence of learning. But this makes it perplexing why the MO learning rate in the volatile condition is slightly positive and slightly greater than in the stable condition.*
>
> *- The pure-EU and pure-MO strategies are interpreted as being analogous to the healthy control group and the patient group, respectively. However, the actual difference in*

estimated EU/MO weighting between the two participant groups was much more moderate. It's unclear whether the same result would be obtained for a more empirically plausible difference in EU/MO weighting.

- The fits of the FLR model to the simulated data "controlled all parameters except for the learning rate parameters across the two strategies" (line 522). If this means that no parameters except learning rate were allowed to differ between the fits to the pure-EU and pure-MO synthetic data sets, the models would have been prevented from fitting the difference in terms of the relative weighting of probability and magnitude, which better corresponds to the true difference between the two strategies. This could have interfered with the estimation of other parameters, such as learning rate.

- If, after addressing all of the above, the FLR model really does recover a spurious difference in learning rate between stable and volatile blocks, it would be worth more examination of why this is happening. For example, is it because there are more opportunities to observe learning in those blocks?

I would recommend performing a version of the Figure 5 simulations using two sets of MOS-model parameters that are identical except that they use healthy-control-like and patient-like values of the EU and MO weights (similar to the parameters described on lines 346-353, though perhaps with the habit controller weight equated). Then fit the simulated data with the FLR model, with learning rate and other parameters free to differ between groups. The result would be informative as to (1) whether the FLR model still misidentifies between-group strategy differences as learning rate differences, and (2) whether the FLR model still identifies spurious learning rate differences between stable and volatile conditions in the control-like group, which become attenuated in the patient-like group.

Many thanks for this great advice. Following your suggestions, we now conduct simulations using the median of the fitted parameters. The representations for healthy controls and patients have identical parameters, except for the three preference parameters; moreover, the habit weights are not controlled to be equal. 20 simulations for each representative, each comprising 4 task sequences sampled from the behavioral data. In this case, we could create error bars and perform statistical tests. We found that the differences in learning rates between stable and volatile conditions, as well as the learning rate adaptation differences between healthy controls and patients, still persisted.

Combined with the discussion in Point 3.2, we justify why a mixture-of-strategy can account for learning rate adaptation as follow. Due to (unknown) differences in task sequences, the MOS6 model exhibits more MO-like behaviors due to the usage of the EU strategy. To capture this behavior pattern, the FLR22 model has to increase its weighting parameter 1-λ for magnitude, which could ultimately drive the FLR22 to adjust the fitted learning rate parameters, exhibiting a learning rate adaptation effect. Our simulations suggest that estimating learning rate just by model fitting may not be the only way to interpret the data.

Action: We included the simulation details in the method section (lines 381-lines 391)

“In one simulated experiment, we sampled the four task sequences from the real data. We simulated 20 experiments with the parameters of

$\beta = 10.803,\ \alpha_{HA} = 0.423,\ \alpha_{\psi} = 0.473,\ w_{EU} = 0.60,\ w_{MO} = 0.15,\ w_{HA} = 0.25$ to

mimic the behavior of the healthy control participants. The first three are the median of the fitted parameters across all participants; the latter three were chosen to approximate the strategy preferences of real health control participants (Figure 4A). Similarly, we also simulated 20 experiments for the patient group with the identical values of β, α_{HA}, and α_{ψ}, but different strategy preferences $w_{EU} = 0.15, w_{MO} = 0.60, w_{HA} = 0.25$. In other words, the only difference in the parameters of the two groups is the switched w_{EU} and w_{MO}. We then fitted the FLR22 to the behavioral data generated by the MOS6 and examined the learning rate differences across groups and volatile contexts (Fig. 6). ”

> *Point 3.4*
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> *Figure 4C shows that the habit-only strategy is able to learn and adapt to changing contingencies, and some of the interpretive discussion emphasizes this. (For instance, line 651 says the habit strategy brings more rewards than the MO strategy.) However, the habit strategy doesn't seem to have any mechanism for learning from outcome feedback. It seems unlikely it would perform better than chance if it were the sole driver of behavior. Is it succeeding in this example because it is learning from previous decisions made by the EU strategy, or perhaps from decisions in the empirical data?*

Yes, the intuition is that the HA strategy seems to show no learning mechanism. But in reality, it yields a higher hit rate than MO by simply learning from previous decisions made by the EU strategy. We run simulations to confirm this (Figure 4B).

> *Point 3.5*
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> *For the model recovery analysis (line 567), the stated purpose is to rule out the possibility that the MOS model always wins (line 552), but the only result presented is one in which the MOS model wins. To assess whether the MOS and FLR models can be differentiated, it seems necessary also to show model recovery results for synthetic data generated by the FLR model.*

Sure, we conducted a model recovery analysis that include all models, and it demonstrates that MOS and FLR can be fully differentiated. The results of the new model recovery analysis were shown in Fig. 7.

> *Point 3.6*
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> *To the best of my understanding, the MOS model seems to implement valence-specific learning rates in a qualitatively different way from how they were implemented in Gagne et al., 2020, and other previous literature. Line 246 says there were separate learning rates for upward and downward updates to the outcome probability. That's different from using two learning rates for "better"- and "worse"-than-expected outcomes, which will depend on both the direction of the update and the valence of the outcome (reward or shock). Might this relate to why no evidence for valence-specific learning rates was found even though the original authors found such evidence in the same data set?*

Thanks. Following the suggestion, we have corrected our implementation of valence-specific learning rate in all models (see lines 261-268).

“To keep consistent with Gagne et al., (2020), we also explored the valence-specific learning rate,

$$\alpha_{\psi} = \begin{cases} \alpha_{\psi+}, & for\ (O(s_1) - \psi(s_1)) > 0 \\ \alpha_{\psi-}, & for\ (O(s_1) - \psi(s_1)) < 0 \end{cases} \quad (6)$$

$\alpha_{\psi+}$ is the learning rate for better-than-expected outcome, and $\alpha_{\psi-}$ for worse-than-expected outcome. It is important to note that Eq. 6 was only applied to the reward context, and the definitions of "better-than-expected" and "worse-than-expected" should change accordingly in the aversive context, where we defined $\alpha_{\psi+}$ for $\left(O(s_1) - \psi(s_1)\right)$ and $\alpha_{\psi-}$ for

$\left(O(s_1) - \psi(s_1)\right) > 0$.

No main effect of valence on learning rate was found (see Supplemental Information Note 3)

> *Point 3.7*
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> *The discussion (line 649) foregrounds the finding of greater "magnitude-only" weights with greater "general factor" psychopathology scores, concluding it reflects a shift toward simplifying heuristics. However, the picture might not be so straightforward because "habit" weights, which also reflect a simplifying heuristic, correlated negatively with the psychopathology scores.*

Thanks. In contrast the detrimental effects of "MO", "habit" is actually beneficial for the task. Please refer to Point 1.12.

> *Point 3.8*
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> *The discussion section contains some pejorative-sounding comments about Gagne et al. 2020 that lack clear justification. Line 611 says that the study "did not attempt to connect the decision process to anxiety and depression traits." Given that linking model-derived learning rate estimates to psychopathology scores was a major topic of the study, this broad statement seems incorrect. If the intent is to describe a more specific step that was not undertaken in that paper, please clarify. Likewise, I don't understand the justification for the statement on line 615 that the model from that paper "is not understandable" - please use more precise and neutral language to describe the model's perceived shortcomings.*

Sorry for the confusion. We have removed all abovementioned pejorative-sounding comments.

> *Point 3.9*
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> *4. Minor suggestions:*
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> *- Line 114 says people with psychiatric illness "are known to have shrunk cognitive resources" - this phrasing comes across as somewhat loaded.*

Thanks. We have removed this argument.

> *- Line 225, I don't think the reference to "hot hand bias" is correct. I understand hot hand bias to mean overestimating the probability of success after past successes. That's not the same thing as habitual repetition of previous responses, which is what's being discussed here.*

Response: Thanks for mentioning this. We have removed all discussions about “hot hand bias”.

> *- There may be some notational inconsistency if alpha_pi on line 248 and alpha_HA on line 253 are referring to the same thing.*

Thanks! Fixed!

> *- Check the notation on line 285 - there may be some interchanging of decimals and commas.*

Thanks! Fixed!

> *Also, would the interpretation in terms of risk seeking and risk aversion be different for rewarding versus aversive outcomes?*

Thanks for asking. If we understand it correctly, risk seeking and risk aversion mechanisms are only present in the RS models, which show clearly worse fitting performance. We thus decide not to overly interpret the fitted parameters in the RS models.

> *- Line 501, "HA and PAT groups" looks like a typo.*
>
> *- In Figure 5, better graphical labeling of the panels and axes would be helpful.*

Response: Thanks! Fixed!

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